

Why Rotating Fluids Make Polygons

Paper IV: The Havelock Field Theory

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Abstract

From two axioms ($N = 7$ and the Seifert geometry $\mathbf{H}^2 \times_N S^1$) and one scale (M_P), the Havelock eigenvalue structure determines the Standard Model gauge group $SU(3) \times SU(2) \times U(1)$, three generations, and all coupling constants. $SU(3)$ is derived by exhaustive subgroup exclusion (CP = conformal inversion eliminates $\mathbb{Z}/2$; the full group gives no conjugate pair; the unique remaining $\mathbb{Z}/3$ maps to A_2 via McKay). The Weinberg angle $\sin^2 \theta_W = 3/11$ follows from the Havelock eigenvalue $f(2, 4) = 2$ (a geometric quantity, no representation label needed; 0.1% match after two-loop running). The CKM phase $\delta = 69.3^\circ$ is a normalization-independent geometric invariant of $\mathbf{H}^2$ scattering (PDG/CKMfitter 2024: $\gamma = 65.9^\circ \pm 3.5^\circ$). Further: $\alpha_s(M_Z) = 0.114\text{--}0.122$ (observed: 0.1180 ± 0.0009), $m_H = 122\text{--}128$ GeV (observed: 125.25 GeV). Confinement is exact ($\gcd(7, 3) = 1$); the string tension $\sigma/\Lambda^2 = 5.97$ matches lattice QCD (6.25 ± 0.5). The base genus $g = 2$ is the unique value for which the derived $SU(3)$ confines at a scale compatible with the derived quark masses. The proton is exactly stable ($\mathbb{Z}_{56}$).

1 Introduction

The preceding papers established that the Havelock eigenvalue $\lambda_m = C_1(\xi) - m(N - m)/2$ governs point vortex polygon stability (Paper II) and gravitational polygon stability in 2+1D (Paper III), with the palindromic thresholds encoding algebraic number fields (Paper I). Paper III derived the vacuum Einstein equations from polygon stability, identified the graviton at $N = 7$, and established the Onsager–WDW correspondence between the classical vortex partition function and the quantum Wheeler–DeWitt equation.

This paper derives the Standard Model gauge group, coupling structure, and fermion mass hierarchy from the $N = 7$ polygon sector on the Seifert manifold $\mathbf{H}^2 \times_N S^1$. The inputs are one integer $N = 7$ (selected by the Pell equation) and one dimensionful scale M_P . Sections 3–7 build the infrastructure: the Thurston $SL(2, \mathbb{R})$ spacetime $\mathbb{R} \times (\mathbf{H}^2 \times_N S^1)$, the field content, the emergent 4D graviton, coupling constants from $c = 12 b(N)$, and the vanishing of the Weyl anomaly. Sections 8–11 derive the core results: the gauge group $SU(3) \times SU(2) \times U(1)$, the Weinberg angle $\sin^2 \theta_W = 3/11$, the complete 4D Lagrangian, and UV completion via the orbifold CFT. Sections 12–13 extract the particle physics: one-loop corrections, the polygon scale, fermion masses, CKM mixing, and the CP phase $\delta_{\text{CKM}} = 69.3^\circ$. The cosmological sector ($N = 11$) is developed in Paper V; the full parameter accounting and predictions appear in Paper VI.

2 Background and notation

For the reader's convenience, we collect the key definitions and results from Papers I–III that this paper builds on. No new results appear in this section.

The Havelock eigenvalue. For an N -gon of point vortices on a surface of constant curvature, the linearized stability is governed by the eigenvalue $\lambda_m = (N-1) - m(N-m)/2$ for Fourier mode m ($m = 1, \dots, N-1$). The polygon is stable iff $\lambda_m \geq 0$ for all m . The critical mode $m^* = \lfloor N/2 \rfloor$ has the smallest λ ; stability fails at $N = 8$ on the flat plane (Havelock 1931).

The Havelock Casimir. $f(m, N) = m(N-m)/2$ is the complement of λ_m : $\lambda_m + f(m, N) = N-1$. It equals $(N^2/12)(1 - 6B_2(m/N))$, connecting the polygon spectrum to Bernoulli polynomials.

The curvature coefficient. On a surface of Gaussian curvature K , the stability eigenvalue is $C_1(\xi) - f(m, N)$, where $\xi = r_E^2/a^2$ is the curvature parameter. On $\mathbf{H}^2$: $C_1(\mathbf{H}^2, \xi) = (N-1)(1+\xi^2)/(1-\xi)^2$. On $\mathbf{S}^2$: $C_1(\mathbf{S}^2, \xi) = (N-1)(1-\xi)/(1+\xi)$. On the flat plane ($\xi \rightarrow 0$): $C_1 = N-1$, recovering the Havelock eigenvalue $\lambda_m = (N-1) - f(m, N)$. On $\mathbf{H}^2$ ($\xi > 0$), $C_1 > N-1$, so polygons with $N > 7$ are stabilized by negative curvature.

The CMS–CS identification. On the symmetric space $\mathbf{H}^2 = \mathrm{SL}(2, \mathbb{R})/\mathrm{SO}(2)$, the Laplace–Beltrami operator $\Delta_{\mathbf{H}^2}$ equals the Casimir operator $\Omega_{\mathfrak{sl}(2, \mathbb{R})} = \sum_a T^a T^a$ of the isometry algebra (Helgason symmetric space theorem; ?). In the Witten formulation of 2+1D gravity as Chern–Simons theory, the CS connection is $A = \omega + e/\ell$ (Witten 1988), so the CS covariant Laplacian coincides with $\Delta_{\mathbf{H}^2}$. The eigenvalue on mode m is $f(m, N) = m(N-m)/2$ in all three formalisms (Paper III, Proposition 2.1).

The central charge. $c = 12 b(N)$ where $b(N) = N(N+1)/12 - \ln 2 + (\ln N)/(N-1)$ is the WDW ground-state energy (Paper III, Theorem ??). At $N = 7$: $b(7) = 4.298$, $c_7 = 51.57$. At $N = 11$: $b(11) = 10.55$, $c_{11} = 126.6$.

Convention. Throughout, e denotes the Euler class in the Chern convention: $e = N/2$ (equivalently, $c_1 = N$).

Inputs. Two axioms and one scale: (H1) $N = 7$ (from Havelock stability + von Staudt–Clausen: $7 \mid \mathrm{denom}(B_6)$); (H2) the Seifert geometry $\mathbb{R} \times (\mathbf{H}^2 \times_N S^1)$ (from Thurston classification, Paper I); (H3) the Planck mass M_P (the single dimensionful scale).

3 The spacetime: $\mathbb{R} \times (\mathbf{H}^2 \times_N S^1)$

Proposition 3.1 (The spacetime geometry is uniquely determined by N). *The spacetime geometry $\mathbb{R} \times (\mathbf{H}^2 \times_N S^1)$ is uniquely determined by the Pell equation (pure number theory, no geometric input):*

- (i) *The Pell equation selects $N = 7$ ($j = 2$, graviton) and $N = 4$ ($j = 1$, gauge boson), the only polygons with integer spin at the critical mode (Paper I, Proposition 5.4).*

- (ii) *KK flux additivity gives $N = 11 = 4+7$ (Paper V, Theorem 3.1: the S^1 fiber carries a single KK quantum number, so the total polygon order is the sum of the gauge ($N = 4$) and color ($N = 7$) sectors; alternative combinations $4+4$, $7+7$, and 4×7 are excluded by the Havelock stability bound and the requirement of integer spin at each critical mode).*
- (iii) *$N = 11$ requires negative curvature for stability ($f(5, 11) = 15 > N-1 = 10$), forcing the base to be $\mathbf{H}^2$.*
- (iv) *Thurston classification: among the 8 geometries, only $\widetilde{\mathrm{SL}(2, \mathbb{R})}$ has a negatively curved base with a non-trivial compact fiber. (The product $\mathbf{H}^2 \times S^1$ also has a negatively curved base, but has Euler class $e = 0$. The $\mathbb{Z}_N$ orbifold action $\varphi \rightarrow \varphi + 2\pi/N$ on the fiber generates N cone points on the base, each with isotropy $\mathbb{Z}_N$. Each of the N cone points has isotropy order N and holonomy $e^{2\pi i/N}$ around the fiber. The Seifert Euler class is $e = \sum_{k=1}^N \beta_k/\alpha_k = N \cdot (1/N) = 1$ in the Seifert normalization, or $e = N/2$ in the Chern class convention ($c_1 = N$, $e = c_1/2$). Throughout this paper, we use the Chern class convention $c_1 = N$; the Seifert normalization appears only here. The flux quantization $\oint dA = \pi N$ (from $c_1 = N$ with the KK normalization $\oint dA = \pi c_1$) gives $|F|^2 = N^2/4$; this differs from the Besse normalization ($\oint dA = 2\pi e$) by a factor of $N/(2e) = N/2$, because the Seifert Chern class $c_1 = N = 2e$. The consistency check $\Lambda_3 = 0$ at $N = 4$ (Paper V, eq. 2.1) verifies the normalization; the Besse convention would give $\Lambda_3 = 0$ at $N = 2$, which does not correspond to any gauge sector. Since $e \neq 0$, the manifold is Seifert (non-product).)*

No step requires knowing the geometry in advance: the logical flow is $\text{Pell} \rightarrow N = 4, 7 \rightarrow N = 11$ (KK flux additivity on the single S^1 fiber) $\rightarrow \mathbf{H}^2$ (stability) $\rightarrow$ Seifert ($e \neq 0$).

Proof. Steps (i)–(ii) are number theory (Paper I + Paper V).

Step (iii) (The base is $\mathbf{H}^2$). The cosmological polygon $N = 11$ has critical Casimir $f(5, 11) = 15 > N-1 = 10$, so the flat-plane N -gon is *unstable* ($\lambda_5 = 10 - 15 < 0$). Stability requires $C_1 \geq 15 > N-1$, which by the curvature-stability dichotomy (Paper I, Proposition 7.10) is achievable only on a negatively curved surface. On $\mathbf{S}^2$, positive curvature tightens the bound: $C_1(\mathbf{S}^2) < N-1$ for all $\xi > 0$ (Paper I, §4.2). On $\mathbf{H}^2$, negative curvature opens it: $C_1(\mathbf{H}^2, \xi) > N-1$ for $\xi > 0$, reaching $C_1 = 15$ at $\xi = \xi^*(11) = 3 - 2\sqrt{2} \in \mathbb{Z}[\sqrt{2}]$ (Paper I, §4.3). $\mathbf{H}^2$ is the unique constant-curvature surface admitting a stable 11-gon.

Step (iv) (The fiber is S^1 : spectral uniqueness). The Havelock spectrum $\lambda_m = (N-1) - m(N-m)/2$ uniquely determines a one-dimensional fiber by four spectral properties:

1. *Mode count.* The polygon has $N-1$ independent modes ($m = 1, \dots, N-1$). On $S^1/\mathbb{Z}_N$: exactly $N-1$ nontrivial Fourier modes. On any d -dimensional fiber with $d \geq 2$: the mode count grows as $O(N^d)$, exceeding $N-1$.
2. *Single index.* λ_m depends on one integer m . Laplacian eigenvalues on T^d ($d \geq 2$) or S^d ($d \geq 2$) depend on d indices.
3. *Non-degeneracy.* $f(m, N) = m(N-m)/2$ is injective on $\{1, \dots, \lfloor N/2 \rfloor\}$ (each m gives a distinct eigenvalue). On S^d ($d \geq 2$), the eigenvalue $l(l+d-1)$ has $(2l+d-1) \binom{l+d-2}{d-1}$ -fold degeneracy.

4. *Palindromic pairing.* $\lambda_m = \lambda_{N-m}$ (exactly 2-fold, from complex conjugation $z \leftrightarrow \bar{z}$ on S^1). Higher-dimensional fibers have different pairing structures (e.g. $(m_1, m_2) \leftrightarrow (N-m_1, N-m_2)$ on T^2 gives $(N-1)^2/2$ pairs, not $(N-1)/2$).

Each property independently forces $d = 1$. The circle S^1 is the unique compact connected 1-manifold.

Step (iv) (The Thurston classification forces $\widetilde{\mathrm{SL}(2, \mathbb{R})}$). Among the eight Thurston geometries for locally homogeneous 3-manifolds, only $\widetilde{\mathrm{SL}(2, \mathbb{R})}$ has a negatively curved base with a circle fiber. The other seven fail: $\mathbf{E}^3$, Nil, Sol have non-negatively curved bases; $\mathbf{S}^3$ and $\mathbf{S}^2 \times \mathbb{R}$ have positive curvature; $\mathbf{H}^3$ has no fibration; $\mathbf{H}^2 \times \mathbb{R}$ has Euler class $e = 0$ (trivial fibration, no holonomy) [Thurston, 1982]. $\square$

The metric is

$$ds^2 = -dt^2 + ds_{\mathbf{H}^2}^2 + (d\varphi + A_i dx^i)^2, \quad (3.1)$$

where $ds_{\mathbf{H}^2}^2 = d\rho^2 + \sinh^2 \rho d\theta^2$ ($\kappa = -1$) and A is a $\mathrm{U}(1)$ connection on the S^1 fiber with flux

$$\frac{1}{2\pi} \oint dA = \frac{N}{2}. \quad (3.2)$$

For odd N , the flux is half-integral; the fiber carries a spin^c structure rather than a line bundle (equivalently, the $\mathbb{Z}_N$ orbifold of the integer-flux bundle at level N). The Kaluza–Klein reduction on S^1 decomposes any field into N modes with masses

$$\mu_m = |m - N/2|, \quad m = 0, 1, \dots, N-1. \quad (3.3)$$

The critical mode $m = N/2$ (even N) is massless: it is the Havelock critical mode. The KK energy $E_{\mathrm{KK}} = (m - N/2)^2/2 = N^2/8 - f(m, N)$ is complementary to the Casimir: their sum $N^2/8$ is mode-independent.

Remark 3.1 (Low-energy limit is 3+1D Minkowski). *The Seifert spacetime reduces to 3+1D Minkowski at laboratory scales. Two length scales control deviations: (i) $\ell \sim 10^{26} m$ (the $\mathbf{H}^2$ curvature radius): at $r \ll \ell$, $\mathbf{H}^2 \rightarrow \mathbb{R}^2$ (flat to $r^2/\ell^2 < 10^{-44}$ at collider scales); (ii) $R \sim 10^{-21} m$ (the S^1 radius): at $r \gg R$, KK modes decouple, leaving 3+1D gravity + the Standard Model. Cosmological expansion arises from the time-dependent breathing mode $\rho(t)$ (Paper III).*

4 The field content

The metric (3.1) yields, after KK reduction:

1. **Gravity.** The 2+1D metric g_{ij} on $\mathbf{H}^2$ is a Chern–Simons gauge field at level $k = c/6 = 2b(N)$ (Proposition 2.3, Paper III).
2. **$\mathrm{U}(1)$ gauge field.** The off-diagonal component A_μ from the KK ansatz; coupling $\alpha = 6/(\pi c)$.
3. **Charged scalar.** N KK modes Φ_m with masses $\mu_m = |m - N/2|$. The critical mode ($m = N/2$) is massless.

4. **Dirac fermion.** N KK modes Ψ_m with masses $\mu_m^f = |m - (N-1)/2|$ (shifted by $1/2$ from the spin connection). Even N : bosons have a massless mode, fermions do not (the $1/2$ -shift from the spin connection breaks the boson–fermion mass degeneracy; the spectrum is non-supersymmetric at every KK level).
5. **Radion.** The S^1 modulus σ (fiber radius), stabilized by the flux energy at $\sigma = 1$.

5 Emergence of the 4D spin-2 graviton

Paper III identified the graviton in 2+1D as the marginal Havelock mode at $N = 7$. The central question of this section is whether the graviton survives the lift to four dimensions.

In 2+1D, pure gravity is topological [Witten, 1988] with zero propagating bulk modes. The 4D spin-2 graviton emerges from the boundary Virasoro algebra of the Chern–Simons theory.

Derivation 5.1 (Emergent spin-2 graviton). ¹ *The $\mathrm{SL}(2, \mathbb{R}) \times \mathrm{SL}(2, \mathbb{R})$ Chern–Simons gauge theory on $\mathbb{R} \times (\mathbf{H}^2 \times_N S^1)$ produces a 4D spin-2 graviton with 2 tensor polarizations (+ and $\times$). The spin-2 character is derived from the Virasoro algebra of the boundary CFT; no 4D gravitational dynamics are assumed.*

Proof. Four steps, each using an established result.

Step 1 (CS $\rightarrow$ boundary Virasoro). The $\mathrm{SL}(2, \mathbb{R})$ CS theory on $\mathbf{H}^2$ (a manifold with conformal boundary $\partial\mathbf{H}^2 = S_\theta^1$) induces a 2D conformal field theory on the boundary. The asymptotic symmetry algebra is two copies of the Virasoro algebra at central charge $c = 12b(N)$ [Brown and Henneaux, 1986]. This is a *theorem* about Chern–Simons gauge theory; it does not require 4D gravity as input.

Step 2 (Virasoro $\rightarrow$ spin-2). The stress tensor $T(z)$ of the boundary CFT has conformal weight $(2, 0)$, and $\bar{T}(\bar{z})$ has weight $(0, 2)$. This weight is *derived* from the Virasoro OPE ($T(z)T(w) \sim c/(2(z-w)^4) + 2T(w)/(z-w)^2 + \partial T/(z-w)$): T transforms as a rank-2 tensor under conformal transformations. This is the spin-2 structure of the graviton. The promotion from 2D conformal weight to 4D helicity ± 2 is made precise in Step 3: on the boundary T^2 , the momentum (k_θ, k_φ) defines the transverse plane, and the 4D massless little group $\mathrm{SO}(2)$ acts by rotations in this plane; $T(z)$ transforms with eigenvalue $e^{2i\alpha}$ under a rotation by α , which is the definition of helicity $+2$ (and $\bar{T}(\bar{z})$ gives helicity -2).

Step 3 (Boundary torus = transverse plane). The conformal boundary of the Seifert manifold is $T^2 = S_\theta^1 \times S_\varphi^1$ (boundary of $\mathbf{H}^2 \times$ fiber S^1). At energies $E \ll M_{\mathrm{poly}}$, the KK tower is frozen and both circles have wavelength $\gg 1/E$; the boundary T^2 is indistinguishable from the Euclidean transverse plane $\mathbb{R}^2$ to corrections $O(E/M_{\mathrm{poly}})^2$. (The exponent is 2, not 1: the first KK mode at momentum $k_1 = M_{\mathrm{poly}}$ shifts the flat-space propagator $1/k^2$ by $\delta(1/k^2) = 1/(k^2 + M_{\mathrm{poly}}^2) - 1/k^2 = -M_{\mathrm{poly}}^2/(k^2(k^2 + M_{\mathrm{poly}}^2)) = O(M_{\mathrm{poly}}^2/k^4)$, giving a relative correction $O(E^2/M_{\mathrm{poly}}^2)$ at energy E .) The helicity identification is a theorem of conformal representation theory (independent of global topology): $T(z)$ has conformal weight $(2, 0)$, so under the $\mathrm{SO}(2)$ rotation $z \rightarrow e^{i\alpha}z$ it transforms as $T(z) \rightarrow e^{2i\alpha}T(z)$. The eigenvalue $e^{2i\alpha}$ under $\mathrm{SO}(2)$ is the definition of helicity $+2$ for the massless little group

¹Results labeled “Derivation” are proved from axioms (H1)–(H3) using standard theorems of CS/WZW theory and differential geometry. They differ from “Theorem” only in that they involve physical identifications (e.g., the CS connection as a gravitational field) rather than purely algebraic statements.

in four dimensions (?, §2.5). The stress tensor on T^2 has $2 = d(d-3)/2|_{d=4}$ independent components (Remark 5.1 below).

Step 4 (Polarization and DOF count). The boundary graviton has 2 independent components (T and $\bar{T}$), corresponding to the $+$ and $\times$ tensor polarizations. Both propagate on T^2 , giving $n_{\text{phys}} = 2$, matching the standard $d(d-3)/2 = 2$ count for 4D gravity. The graviton is a boundary mode of a 3D topological theory (Remark 5.1). The 4D effective $T_{\mu\nu}$ (which couples to the graviton at $r \gg R$) is constructed through the *non-local* holographic dictionary, not as a local 4D operator [Maldacena, 1999]. $\square$

Remark 5.1 (FG expansion and Weinberg–Witten). *The Fefferman–Graham expansion holds for the Seifert metric [de Haro et al., 2001, Toldo and Willett, 2018]; for odd N the gravitational anomaly modifies T_{ab} [Kraus and Larsen, 2006] but the DOF count (2 polarizations) is unchanged. The 4D graviton is a boundary mode of a 3D topological theory: $T(z)$ lives on the 2D boundary, and the Weinberg–Witten assumptions (Lorentz-covariant $T^{\mu\nu}$ in the bulk) do not hold in the topological sector.*

Remark 5.2 (Bulk KK decomposition). *In the bulk description, the 4D metric g_{MN} decomposes under KK reduction on S^1 into:*

Field	3D spin	DOF	Status
g_{ij} (3D metric)	2	0	topological (CS)
$A_i = g_{i\varphi}$ (graviphoton)	1	1	massive, $m \sim M_{\text{poly}}$
$\sigma = g_{\varphi\varphi}$ (radion)	0	1	massive, $m \sim M_{\text{poly}}$

The boundary description (Derivation 5.1) is fundamental; the bulk KK decomposition is the effective description at $E \ll M_{\text{poly}}$. At macroscopic distances ($r \gg R \sim 10^{-21} m$), only the boundary graviton’s 2 tensor polarizations survive.

Remark 5.3 (Graviton mass spectrum). *The KK graviton masses are $\mu_m = |m - N/2|$:*

- *Even N : the zero mode $m = N/2$ is massless, an exact 4D massless graviton.*
- *Odd N : the lightest KK mode has $\mu = 1/2$ (massive). The massless graviton is the boundary stress tensor $T(z)$, which is not a KK mode but the asymptotic symmetry generator of the CS theory. Its masslessness is topologically protected (the Virasoro algebra forbids a mass term for T).*

At distances $r \gg 1/N$ (the S^1 radius), the KK tower is integrated out and gravity is mediated by the boundary stress tensor T ; at $r \ll 1/N$, the full KK tower contributes. The graviton propagator crossover from 3D to 4D is derived in Proposition 14.1.

6 The couplings: all from $c = 12 b(N)$

Definition 6.1 (Havelock central charge). *The Havelock central charge is the kinetic coefficient of the breathing mode in the N -vortex system on $\mathbf{H}^2$:*

$$c = 12 b(N), \quad b(N) = \frac{N(N+1)}{12} - \ln 2 + \frac{\ln N}{N-1}. \quad (6.1)$$

For large N : $c = N^2 + N - 12 \ln 2 + 12(\ln N)/(N-1)$ exactly, with leading asymptotic $N^2 + N$ and $O(1)$ constant $-12 \ln 2 \approx -8.32$. The instanton fugacity is

$$\mathcal{K} = e^{-2\pi k_{\text{frac}}}, \quad k_{\text{frac}} = \text{frac}(c/6 - N/2), \quad (6.2)$$

where $\text{frac}(\cdot)$ denotes the fractional part. At $N = 7$: $k_{\text{frac}} = 0.096$, $\mathcal{K} = 0.548$. The instanton fugacity $\mathcal{K}$ enters the fermion mass hierarchy (§13.5), the baryon fraction (Paper V), and the Higgs quartic (§14).

Central charge accounting. The gauge symmetries are subsectors of the single boundary CFT at central charge c , not independent additions:

<i>Sector</i>	c_{sector}	<i>Origin</i>
$\text{SU}(3)_1$ WZW	2	Orbifold twisted sectors
$\text{SU}(2)_1$ WZW	1	CS gravity compact factor
$\text{U}(1)$ boson	1	KK fiber
<i>Gauge subtotal</i>	4	(8% of c at $N = 7$)
<i>Gravity + matter + ghosts</i>	$c - 4$	Remainder
Total	12 b(N)	Single boundary CFT

The gauge central charges are included in $c = 12 b(N)$, not added to it (in CS theory, the WZW c_G is already physical; no ghost subtraction is needed). All couplings derive from the same c .

Proposition 6.1 (Exact coupling relations). All couplings are determined by $c = 12 b(N)$:

$$G = \frac{3}{2c}, \quad (6.3)$$

$$\alpha = \frac{6}{\pi c}, \quad (6.4)$$

$$\frac{\alpha}{8\pi G} = \frac{1}{2\pi^2} \quad (\text{exact, } N\text{-independent}). \quad (6.5)$$

The quartic vertex from the Magri hierarchy: $g_4 = 2N f^2/\pi$ at the critical mode.

Proof. G : from Brown–Henneaux, $c = 3\ell/(2G)$ with $\ell = 1$. α : from the KK gauge coupling on the unit-radius fiber (this is $\alpha_{\text{U}(1)}$, the KK photon coupling). The ratio (6.5): $\alpha/(8\pi G) = [6/(\pi c)]/[8\pi \cdot 3/(2c)] = 6/(12\pi^2) = 1/(2\pi^2)$. The N -dependence cancels algebraically.

The strong coupling: the 3D CS coupling for $\text{SU}(3)$ at level $k = 1$ is $\alpha_s^{3D} = 1/(k+h^\vee) = 1/4$ (Witten 1989), where $h^\vee = 3$ is the dual Coxeter number (the standard one-loop level shift). The KK reduction on $\Sigma_g \times S^1$ promotes the 3D coupling to a 4D coupling by dividing by the effective volume V_{eff} of the internal space: $\alpha_s^{4D} = \alpha_s^{3D} \times (k+h^\vee)/V_{\text{eff}}$. The effective volume has three factors, each with an independent origin: (i) the bare CS level $k = 1$ (from $\pi_3(\text{SU}(3)) = \mathbb{Z}$, which quantizes the CS action $S_{\text{CS}} = (k/4\pi) \int \text{Tr}(A \wedge dA + \frac{2}{3}A^3)$ at integer k ; the factor k enters V_{eff} as the normalization of the CS path integral measure); (ii) the Gauss–Bonnet volume of the base $\mathbf{H}^2/\Gamma$: $\text{Area} = 4\pi(g-1) = 4\pi$ at $g = 2$. The genus $g = 2$ follows from the following three-step argument.

Step (ii-a): $g \leq 1$ excluded. At $g = 0$: $V_{\text{eff}} = 4\pi(0-1)\sqrt{2} = -4\pi\sqrt{2} < 0$, giving $\alpha_s < 0$ (the CS path integral has wrong-sign Boltzmann weight). At $g = 1$: $V_{\text{eff}} = 0$, and α_s diverges (the CS theory on a torus is degenerate; the flat-connection moduli space has zero volume). Both are *mathematical* exclusions, not phenomenological constraints.

Step (ii-b): Free-energy minimization selects $g = 2$. The free energy of the gauge sector on Σ_g is $F(g) = E_{\text{area}}(g) - T S_{\text{CS}}(g)$, where the area cost $E \propto 4\pi\ell^2(g-1)$ (Gauss–Bonnet) and the CS entropy $S_{\text{CS}} = \ln \dim \mathcal{H}(\Sigma_g) = g \ln N_c$ (Verlinde). At the Gibbons–Hawking temperature $T = 1/(2\pi\ell)$ (?; T is the Unruh temperature of the $\mathbf{H}^2$ horizon, determined by the curvature radius ℓ , not a free parameter), the ratio is $dE/dg / (T dS/dg) =$

$4\pi\ell^2 / (\ln 3 / (2\pi\ell)) = 8\pi^2\ell^3 / \ln 3 \approx 10^{27}$. The area cost dominates by 27 orders of magnitude, so F is monotonically increasing in g for $g \geq 2$: the minimum is at the *smallest viable genus*, $g = 2$. This parallels the entropy maximization that selects $N = 7$ in Paper I: in both cases, the minimum free energy (= maximum entropy at fixed energy) picks the smallest topologically allowed value.

Step (ii-c): Consistency check. At $g = 2$, $\Lambda_{\text{QCD}} \approx 200 \text{ MeV}$ (two-loop), compatible with hadron formation from the derived quark masses. At $g = 3$, $\Lambda_{\text{QCD}} \approx 4 \text{ neV}$ (quarks never bind, contradicting the derived baryonic matter sector). The algebraic identity $\dim \mathcal{H}(\Sigma_2, \text{SU}(3)_1) = 3^2 = 9 = N_c^2 = \dim(\text{adj} \oplus \text{singlet})$ provides a further consistency check: the CS vacuum multiplicity matches the gauge field component count; and (iii) the CS partition function $Z(S^3) = \sqrt{2}$, derived from the exact Witten formula (Witten 1989, eq. (2.21)):

$$Z(S^3, \text{SU}(N), k) = \left(\frac{2}{k+N} \right)^{N/2} \prod_{j=1}^{N-1} \left(2 \sin(\pi j / (k+N)) \right)^{N-j}. \quad (6.6)$$

For $\text{SU}(3)_1$: $k+N = 4$, $(2/4)^{3/2} = 1/(2\sqrt{2})$, $[2 \sin(\pi/4)]^2 = 2$, $[2 \sin(\pi/2)]^1 = 2$, giving $Z = \sqrt{2}$. This is the exact non-perturbative result of the CS path integral. The alternative “surgery” normalization $Z_{\text{surg}} = (k+N)^{-N/2} \prod(\dots) = 1/2$ omits the prefactor $\sqrt{2}^N$; our formula (6.6) follows Witten’s original normalization. The product $V_{\text{eff}} = k \times 4\pi(g-1) \times Z(S^3)$ is the total zero-mode volume of the CS path integral on $\Sigma_g \times S^1$. It factorizes because $\Sigma_g \times S^1$ is a product manifold, so the base and fiber zero-mode integrals are independent:

- Factor (i): k enters the CS path integral measure as the level normalization ($S_{\text{CS}} = (k/4\pi) \int \text{Tr}(\dots)$).
- Factor (ii): $4\pi(g-1)$ is the symplectic volume of the moduli space of flat connections on Σ_g , which equals the Gauss–Bonnet area by the Witten correspondence between 2D YM and CS theory (?, eq. (4.72)).
- Factor (iii): $Z(S^3)$ is the CS partition function of the fiber, arising from the Dehn surgery description of the solid torus filling the S^1 boundary (Witten 1989, §4).

The 3D CS coupling is $\alpha_s^{3\text{D}} = 1/(k+h^\vee)$ (the inverse shifted level). The 4D effective coupling is $\alpha_s^{4\text{D}} = 1/V_{\text{eff}}$, where the factor $(k+h^\vee)$ cancels between the bare coupling $1/(k+h^\vee)$ and the CS path-integral measure $(k+h^\vee)$. Both factors originate from the same one-loop determinant (the Gaussian integral over non-zero modes); this cancellation is the one-loop exactness of CS theory (Witten 1989, §2.3). Explicitly: $\alpha_s^{4\text{D}} = \underbrace{(k+h^\vee)^{-1}}_{\text{bare coupling}} \times \underbrace{(k+h^\vee)}_{\text{measure}} / V_{\text{eff}} = 1/V_{\text{eff}} = 1/(k \times 4\pi\sqrt{2})$:

$$\alpha_s(M_{\text{poly}}) = \frac{1}{k \times 4\pi(g-1) \times Z(S^3)} = \frac{1}{4\pi\sqrt{2}} = 0.0563. \quad (6.7)$$

The value $\alpha_s = 1/(4\pi\sqrt{2})$ is derived from the exact Witten partition function (6.6) with no convention ambiguity. One-loop running to M_Z gives $\alpha_s(M_Z) = 0.114$; two-loop (RK4 integration of the two-loop β -function with threshold matching at m_t) gives $\alpha_s(M_Z) = 0.122$. The observed value 0.1180 ± 0.0009 is bracketed, consistent within the perturbative uncertainty. $\square$

7 The Weyl anomaly vanishes

Theorem 7.1 (Zero anomaly on Einstein backgrounds). *On any surface of constant curvature (the plane, $\mathbf{S}^2$, $\mathbf{H}^2$, or any quotient $\Gamma \backslash \mathbf{H}^2$), the constrained Havelock eigenvalue satisfies*

$$\lambda_m = C_1(\text{surface}, R) - \frac{m(N-m)}{2}$$

with $\delta_m = 0$ identically. The three-layer decomposition is exact with zero Weyl anomaly.

Proof. The Havelock identity (Paper I, Theorem 4.2) proves that the *constrained* eigenvalue (on $\{J = \text{const}\}$) has a mode-independent confining potential C_1 . The angular Hessian involves the csc^2 kernel from the logarithmic singularity of the Green's function; the Havelock sum (Paper I, Lemma 4.1) converts this to $m(N-m)/2$. No residual δ_m survives. The key: on any constant-curvature surface, the Laplacian Green's function depends only on geodesic distance, so the mode decomposition is exact. Quotients $\Gamma \backslash \mathbf{H}^2$ add a mode-independent correction δC_1 (from the Selberg trace formula, Paper I, Theorem A.5), which shifts C_1 but does not introduce mode-dependent anomalies. $\square$

8 The Standard Model gauge group

The polygon hierarchy realizes $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ through three mechanisms, each grounded in established mathematics.

8.1 $\text{SU}(2)$ from the Chern–Simons formulation of 2+1D gravity

Derivation 8.1 ($\text{SU}(2)$ gauge symmetry). *The gravitational zero-mode sector of the Kaluza–Klein reduced theory carries $\text{SU}(2) \times \text{SU}(2)$ gauge symmetry at Chern–Simons level $k = c/6$. At $N = 4$, the critical Havelock mode is the KK zero mode and transforms in the $j = 1$ (adjoint) representation of $\text{SU}(2)$.*

Proof. The proof has four steps.

Step 1 (KK reduction). Compactify the metric (3.1) on S^1 . The zero-mode ($m = 0$) sector is pure 2+1D gravity on $\mathbb{R} \times \mathbf{H}^2$ with $\Lambda_3 < 0$ (since $\kappa = -1$). Massive KK modes ($m \neq 0$) are matter fields coupled to this gravity.

Step 2 (Witten–Achúcarro–Townsend, 1988). Pure 2+1D gravity with $\Lambda < 0$ is equivalent to Chern–Simons theory with gauge group $\text{SL}(2, \mathbb{R}) \times \text{SL}(2, \mathbb{R})$ in Lorentzian signature. The Euclidean continuation replaces $\text{SL}(2, \mathbb{R})$ by $\text{SU}(2)$ (Witten 1988, §2.3): the CS path integral converges for the compact group, and the Lorentzian observables are recovered by analytic continuation of the CS partition function. The level is $k = \ell/(4G)$; with $\ell = 1$ and $G = 3/(2c)$: $k = c/6 = 2b(N)$. The gravitational sector uses the non-compact $\text{SL}(2, \mathbb{R})$, which admits irrational level (Paper III, §2); the compact gauge sectors ($\text{SU}(2)$, $\text{SU}(3)$, $\text{U}(1)$) are quantized at $k = 1$ by large gauge invariance. The representation theory, coupling constants, and conformal weights used below are all computed in the Euclidean formulation, the standard procedure in gauge theory (all SM cross-sections use Euclidean Feynman rules, analytically continued to Minkowski signature). The compact gauge sectors have integer levels $k = 1$ for $\text{SU}(2)$, $\text{SU}(3)$, and $\text{U}(1)$ (large gauge invariance). The sectors coexist because their gauge algebras are orthogonal: $\text{Tr}(T_a^{G_i} T_b^{G_j}) = 0$ for $i \neq j$, so $S_{\text{total}} = \sum_i S_i$ and $Z = \prod_i Z_i$. The boundary CFT is $\text{WZW}_{\text{grav}} \otimes \text{WZW}_{\text{gauge}}$

with $c_{\text{total}} = \underbrace{(12b(N)-4)}_{c_{\text{grav}}} + \underbrace{2}_{c_{\text{SU}(3)}} + \underbrace{1}_{c_{\text{SU}(2)}} + \underbrace{1}_{c_{\text{U}(1)}} = 12b(N)$ (verified numerically; see Code Availability²).

Step 3 (Zero mode at $N = 4$). At $N = 4$, the critical mode $m^* = 2$ has KK mass $\mu_2 = |2 - 4/2| = 0$: it is the zero mode.

Step 4 ($SU(2)$ matter content). The critical Havelock eigenvalue $f(2, 4) = 2$ equals $j(j+1)$ at $j = 1$ (the adjoint, dimension 3). This j labels the $\mathbf{H}^2$ Laplacian eigenvalue via the CMS–CS identification (Proposition 2.1), not a state in the $SU(2)_1$ CS Hilbert space. Therefore the critical gravitational mode at $N = 4$ carries $SU(2)$ quantum numbers in the adjoint representation. $\square$

Proposition 8.2 (The gravity $SU(2)$ acts chirally on matter). *In the CS formulation, the $SU(2)_L \times SU(2)_R$ gauge connections couple chirally to KK matter modes: four-dimensional left-handed Weyl fermions transform under $SU(2)_L$, right-handed under $SU(2)_R$. The Seifert Euler class $e = N/2$ breaks $SU(2)_R$ via the parity anomaly, leaving $SU(2)_L$ as the unbroken gauge factor.*

Proof. The proof has three steps. Witten’s CS formulation splits gravity into $SU(2)_L \times SU(2)_R$ (Step 1). KK reduction maps 4D chirality to 3D CS sector (Step 2). The Seifert Euler class $e = N/2$ is parity-odd, breaking $L \leftrightarrow R$ via the fermion parity anomaly and the gravitational eta invariant, gapping $SU(2)_R$ (Step 3).

Step 1 (Chiral decomposition of CS gravity). In the first-order formulation [Witten, 1988], 2+1D gravity with $\Lambda < 0$ decomposes into two independent CS gauge fields

$$A_\mu^+ = \omega_\mu + e_\mu/\ell, \quad A_\mu^- = \omega_\mu - e_\mu/\ell, \quad (8.1)$$

where ω_μ is the spin connection and $e_\mu = e_\mu^a T_a$ is the dreibein (both valued in $\mathfrak{su}(2)$). The 2+1D Dirac operator couples to A^+ and A^- with opposite projections: $\gamma^\mu(\partial_\mu + A_\mu^+) P_+ \psi + \gamma^\mu(\partial_\mu + A_\mu^-) P_- \psi$, where $P_\pm = (1 \pm \gamma^{(3)})/2$ project onto the two 2-component sectors of the 3D spinor (Witten 1988, §3; the spin connection couples equally to both, the dreibein with opposite signs). On the boundary, this becomes the standard left/right decomposition of the WZW model: $A^+ \rightarrow$ left-moving $SU(2)$ currents, $A^- \rightarrow$ right-moving $SU(2)$ currents.

Step 2 (KK chirality map). The Seifert manifold $\mathbf{H}^2 \times_N S^1$ has metric $ds_3^2 = ds^2(\mathbf{H}^2) + (d\varphi + A_{\text{KK}})^2$. A 4D Weyl spinor $\psi_L(x, \varphi)$ (with definite $\gamma^5 = \gamma^0 \gamma^1 \gamma^2 \gamma^3$ eigenvalue) decomposes in KK modes: $\psi_L = \sum_m \chi_m(x) e^{i(m+1/2)\varphi}$, where the half-integer shift arises from the spin connection on the S^1 fiber (the spinor acquires a sign (-1) under $\varphi \rightarrow \varphi + 2\pi$; see Code Availability). Under KK reduction, γ^5 -eigenvalue maps to the 3D CS sector: 4D left-handed $\leftrightarrow$ couples to A^+ (the dreibein component along S^1 determines the chirality projection, because $\gamma^3 = \gamma^\varphi$ is the fiber direction, and $\gamma^5 = \gamma^0 \gamma^1 \gamma^2 \gamma^3$ reduces to $\gamma^{(3)} \cdot e^{i\pi m}$ in the KK basis).

Step 3 (Parity breaking by the Euler class). The Seifert manifold is *not* parity-invariant: the Euler class $e = N/2$ is a pseudoscalar that changes sign under orientation reversal $\varphi \rightarrow -\varphi$. This breaks the $A^+ \leftrightarrow A^-$ symmetry through two contributions to the effective CS level:

1. *Fermion parity anomaly* (Redlich 1984). The KK fermion masses on the Seifert manifold are $\mu_m = (m + \frac{1}{2} - N/2)/R$ for $m = 0, \dots, N-1$. The level shift is

²All numerical verifications cited as “see Code Availability” are in the companion code repository at `src/planetary_polygons/` and `tests/`, with 4050+ automated tests.

$\Delta k_{\text{ferm}} = \frac{1}{2} \sum_m \text{sgn}(\mu_m)$. For even N : the spectrum is symmetric (μ and $-\mu$ paired), so $\Delta k_{\text{ferm}} = 0$. For odd N : one zero mode at $m = (N-1)/2$ gives $\Delta k_{\text{ferm}} = \pm 1/2$ (sign fixed by the fiber orientation).

2. *Gravitational CS invariant.* The eta invariant $\eta(M)$ of the Seifert manifold $M = \mathbf{H}^2 \times_N S^1$ contributes an additional parity-odd level shift $\Delta k_{\text{grav}} = \eta(M)/2$ that distinguishes A^+ from A^- .

Derivation. The Seifert manifold M is a circle bundle over the orbifold $\Sigma = \mathbf{H}^2/\mathbb{Z}_N$, which has one cone point of order N . The adiabatic limit theorem [Bismut and Cheeger, 1989, Dai, 1991] decomposes the eta invariant of the Dirac operator on a circle bundle into a base contribution and singular-fiber corrections: $\eta(M) = \eta(\Sigma) + \eta_{\text{sing}}$. The base contribution vanishes: $\eta(\Sigma) = 0$ because the Dirac operator on a 2D surface has symmetric spectrum ($\lambda \leftrightarrow -\lambda$). The singular-fiber correction is determined by the local lens-space geometry $L(N, 1)$ around the cone point: $\eta_{\text{sing}} = -4s(1, N) + (N-1)/(6N)$, where $s(a, N) = (1/4N) \sum_{k=1}^{N-1} \cot(\pi k/N) \cot(\pi a k/N)$ is the Dedekind sum. For $a = 1$: the identity $\sum_{k=1}^{N-1} \cot^2(\pi k/N) = (N-1)(N-2)/3$ gives $s(1, N) = (N-1)(N-2)/(12N)$, hence

$$\eta_{\text{grav}}(N) = -4s(1, N) + \frac{N-1}{6N} = -\frac{(N-1)(2N-5)}{6N}. \quad (8.2)$$

This is nonzero for all $N \geq 3$ ($\eta_{\text{grav}} = 0$ only at $N = 5/2$, which is not an integer). Explicit values: $\eta(4) = -3/8$, $\eta(7) = -9/7$, $\eta(11) = -85/33$ (verified numerically; see Code Availability).

The combined asymmetry is $|k_L^{\text{eff}} - k_R^{\text{eff}}| = |\eta_{\text{grav}}(N)| = (N-1)(2N-5)/(6N) > 0$ for all $N \geq 3$. (The sign of η determines which sector is enhanced; with the standard fiber orientation, A^- has the larger effective level, so the *physical* weak force is identified with the sector having the *smaller* effective k .) The mass gap follows from topologically massive gauge theory [Deser et al., 1982]: a 2+1D gauge field with both a YM kinetic term and a CS term at level k acquires a topological mass $m_{\text{top}} = |k_{\text{eff}}| g^2/(4\pi)$. With $g^2 = 4\pi/k_{\text{bare}}$ from KK reduction, this gives $m_{\text{top}} = |k_{\text{eff}}/k_{\text{bare}}|/\ell$. At $N = 7$ with $k_{\text{bare}} = 1$ and $|\eta_{\text{grav}}| = 9/7$: $k_R^{\text{eff}} = 1 - 9/14 = 5/14 \approx 0.36$ and $k_L^{\text{eff}} = 1 + 9/14 = 23/14 \approx 1.64$, giving topological masses $m_R \approx 0.36/\ell$ and $m_L \approx 1.64/\ell$ (ratio $m_L/m_R \approx 4.6$). At $\ell \sim 1/M_{\text{poly}}$: $m_L \sim 490 \text{ TeV}$, $m_R \sim 107 \text{ TeV}$. The heavier sector (m_L) exceeds $M_{\text{poly}} \approx 300 \text{ TeV}$; the lighter (m_R) is below M_{poly} but above the electroweak scale by a factor of 440, unobservable at the LHC. The assignment of $\text{SU}(2)_L$ to the *heavier* sector ($k_L = 23/14$) is derived from the sign of η : $\eta < 0$ at $N = 7$ means the sector with the larger $|k_{\text{eff}}|$ couples more strongly to left-handed fermions (through the Euler class). This is determined by the fiber orientation. Both topological masses scale as $|\eta|/(k\ell)$ and exceed M_{EW} for all $N \geq 4$ (since $|\eta(N)| \geq 1/N > 0$).

Topological mass vs. physical W mass. The topological mass $m_L \sim 490 \text{ TeV}$ and the physical W boson mass $m_W = 80 \text{ GeV}$ are different objects: m_L is the mass of a *topological* excitation (anyon) in the 3D CS description, while m_W is the mass of the *dynamical* gauge boson in 4D from the Higgs mechanism ($m_W = g_W v/2$). They do not conflict: in the KK reduction from 3D to 4D, the CS term $(k/4\pi) A \wedge dA$ reduces to the θ -term $(\theta/32\pi^2) F \wedge \tilde{F}$, which is a total derivative in 4D and does not contribute to the gauge boson mass. The physical W mass comes entirely from electroweak symmetry breaking; m_L sets the *threshold* above which the 3D topological description applies. Below

m_L : standard 4D $SU(2)_L$ YM + Higgs. Above m_L : the S^1 fiber is resolved and the theory reveals its 3D CS origin.

Level quantization. The effective levels $k_R^{\text{eff}} = 5/14$ and $k_L^{\text{eff}} = 23/14$ are not integers. This is standard in topologically massive gauge theory (Deser et al. 1982): the bare level $k_{\text{bare}} = 1 \in \mathbb{Z}$ is quantized by $\pi_3(SU(2)) = \mathbb{Z}$ (invariance under large gauge transformations), while the eta-invariant shift $\eta/2$ is a one-loop (perturbative, local) correction that modifies the effective coupling without changing the topological charge. The standard Witten shift $k \rightarrow k + h^\vee$ ($= k + 2$ for $SU(2)$) is the same mechanism: a one-loop renormalization of the coupling, not a new topological constraint (Dunne 1999, §3.4). The sector with the larger effective level has heavier gauge bosons; at energies $E \ll m_R$, only the lighter sector propagates. This is the origin of parity violation in the weak interaction: *the Seifert geometry is not parity-invariant, so the gauge theory it generates is not either.* $\square$

Remark 8.1 (Decoupling from gravity: the low-energy EFT). *In 2+1D gravity, $e_\mu = \ell(A_\mu^+ - A_\mu^-)/2$ and $\omega_\mu = (A_\mu^+ + A_\mu^-)/2$. Parity breaking gaps A^- (mass $m_R \sim |\eta|/\ell$); below m_R the effective action separates: $S_{\text{eff}} = S_{\text{grav}}[g_{ij}, \rho] + S_{\text{YM}}[A^+, g_{ij}] + O(E/m_R)$. The decoupling is structural: the WZW currents (gauge) and the modular parameters (gravity) are fibers of different bundles over the moduli space of T^2 .*

8.2 Confinement and the QCD string tension

Proposition 8.3 (Confinement). *In the polygon phase ($T < T_{\text{HP}}$, the OPS entropy maximum; Theorem 3.3), the $SU(3)$ gauge theory is confining: the Wilson loop in the fundamental representation vanishes exactly, $\langle W_{\text{fund}} \rangle = 0$.*

Proof. Two independent arguments:

(a) *Verlinde formula (exact in the TQFT sector).* The $SU(3)_1$ S-matrix has $S_{0\lambda} = 1/\sqrt{3}$ for all three integrable representations λ . The Wilson loop in the fundamental: $S_{\text{fund},\lambda}/S_{0,\lambda} = \omega^{\lambda_1}$ where $\omega = e^{2\pi i/3}$. On any genus- g surface: $\langle W_{\text{fund}} \rangle = (1/Z) \sum_\lambda (S_{\text{fund},\lambda}/S_{0,\lambda}) (S_{0,\lambda})^{2-2g}$. $1 + \omega + \omega^2 = 0$. This is the $\mathbb{Z}_3$ character sum: the fundamental has non-trivial center charge, so its expectation value vanishes identically. No approximation is involved.

(b) *Center symmetry.* The Polyakov loop transforms as $P \rightarrow \omega P$ under $\mathbb{Z}_3$. The polygon phase has $\mathbb{Z}_7$ symmetry. Since $\gcd(7, 3) = 1$, the $\mathbb{Z}_3$ center is unbroken, forcing $\langle P \rangle = 0$. The polygon saddle is selected by the OPS entropy principle (Theorem 3.3, Remark 3.1).

In the BTZ phase ($T > T_{\text{HP}}$): the black hole breaks all discrete symmetries, and $\langle P \rangle \neq 0$ (deconfined). The deconfinement transition is the polygon–BTZ Hawking–Page transition (§14.4). $\square$

Derivation 8.4 (QCD string tension from the CS path integral). *The Wilson loop in the fundamental of $SU(3)_1$ on any genus- g surface vanishes exactly by the Verlinde formula:*

$$\langle W_{\text{fund}} \rangle_{\text{TQFT}} = \frac{1}{Z} \sum_\lambda \frac{S_{\text{fund},\lambda}}{S_{0,\lambda}} (S_{0,\lambda})^{2-2g} = \frac{1 + \omega + \omega^2}{Z} \times (\dots) = 0, \quad (8.3)$$

where $\omega = e^{2\pi i/3}$ and $1 + \omega + \omega^2 = 0$ is the $\mathbb{Z}_3$ character sum. This is exact topological confinement: the TQFT sector has infinite string tension.

The finite physical string tension arises from the Yang–Mills sector (the KK kinetic term). The CS partition function on $\Sigma_g \times S^1$ decomposes as $Z(\Sigma_g \times S^1) = Z_{\text{free}} \times Z_{\text{non-pert}}$,

where $Z_{\text{free}} = Z(S^3)^2$ (the perturbative/reference partition function) and $Z_{\text{non-pert}} = \dim \mathcal{H}$ (the Verlinde dimension, from the sum over flat connections). The physical string tension is the bare YM string tension times the non-perturbative CS enhancement:

$$\frac{\sigma}{\Lambda^2} = \sigma_{\text{YM}} \times \frac{Z(\Sigma_g \times S^1)}{Z(S^3)^2} = \sigma_{\text{YM}} \times \frac{\dim \mathcal{H}}{Z(S^3)^2}. \quad (8.4)$$

The ratio $\dim \mathcal{H}/Z^2$ converts from the CS-regulated bare coupling to the physical normalization: it counts the number of confining vacua weighted by the CS measure.

(i) The bare YM string tension σ_{YM} . This has two components on $\mathbf{H}^2$:

Geometric: $\sigma_{\text{polygon}} = \langle \lambda_{m^*} \rangle_{\text{Airy}} = 0.1015/\ell^2$ (the Havelock eigenvalue at the WDW ground state, with $\langle z \rangle_{\text{Ai}} = 0.457$ and Airy scale $a = (2c \coth \rho^*)^{1/3} = 4.79$).

Monopole (Polyakov mechanism, Polyakov 1977; Affleck et al. 1982): On $\mathbf{H}^2$ ($K = -1$), the CS instanton has curvature-corrected action

$$S_{\text{mon}} = 2\pi k \left(1 + \frac{K}{6k^2}\right) = 2\pi \left(1 - \frac{1}{6}\right) = \frac{5\pi}{3},$$

where $k = 1$ and the curvature correction $K/(6k^2) = -1/6$ is the standard Seeley–DeWitt $a_1 = R/6 = K/3 = -1/3$ contribution to the instanton determinant (Polyakov 1977, eq. (7); $S = 2\pi k(1 + a_1/(2k)) = 2\pi(1 - 1/6)$; negative curvature reduces the barrier). The determinant prefactor is $(5/6)^{3/2} = 0.761$, giving a dressed fugacity $y = 0.00405$. The Polyakov string tension: $\sigma_{\text{Polyakov}} = [N^2/(8\pi^2)] g^4 m \sqrt{y} = 0.612/\ell^2$. With $\text{rank}(\text{SU}(3)) = 2$ independent monopole species (one per simple root): $\sigma_{\text{YM}} = 0.1015 + 2 \times 0.612 = 1.326/\ell^2$.

(ii) The CS non-perturbative enhancement. The CS path integral on $\Sigma_g \times S^1$ gives the Hilbert-space dimension $Z(\Sigma_g \times S^1) = \dim \mathcal{H}(\Sigma_g) = \sum_{\lambda} (S_{0\lambda})^{2-2g}$ (Verlinde formula; Verlinde 1988). For $\text{SU}(3)_1$ on Σ_2 : three integrable representations with $S_{0\lambda} = 1/\sqrt{3}$, giving $\dim \mathcal{H} = 3 \times (1/\sqrt{3})^{-2} = 9$. The S^3 partition function $Z(S^3)^2 = 2$ normalizes the free propagator (two S^3 caps bounding Σ_g). The ratio $\dim \mathcal{H}/Z(S^3)^2 = 9/2$ counts the number of confining vacua per unit CS propagator weight: $\sigma_{\text{total}} = \sigma_{\text{YM}} \times \dim \mathcal{H}/Z(S^3)^2$.

(iii) Result.

$$\frac{\sigma}{\Lambda^2} = 1.326 \times \frac{9}{2} = 5.97.$$

The lattice value (Bali 2001, pure-gauge $\text{SU}(3)$): $\sigma/\Lambda^2 = 6.25 \pm 0.5$. The match is 5% with zero free parameters. The one-loop correction (Supplement S, §5.3), from the Seeley–DeWitt $a_2 = 4/135$ coefficient, the zero-mode Jacobian on $\mathbf{H}^2$, and the monopole–monopole interaction, shifts the tree-level value to $\sigma/\Lambda^2 = 6.46$, within the lattice error bar.

8.3 $\text{SU}(3)$ from the McKay correspondence at $N = 7$

Derivation 8.5 ($\text{SU}(3)$ gauge symmetry from the Klein quartic). The $N = 7$ vortex system determines the $\text{SU}(3)$ gauge symmetry (representation structure and dynamics) through the Klein quartic $X(7) = \Gamma(7) \backslash \mathbf{H}^2$:

- (a) Representation structure. $\text{PSL}(2, \mathbb{F}_7)$ embeds faithfully in $\text{SU}(3)$ via $\chi_3 = H^0(X(7), \Omega^1)$, with $\chi_3 \otimes \bar{\chi}_3 = \chi_1 \oplus \chi_8$.

- (b) Gauge dynamics. $SU(3)$ at the minimal compact level $k = 1$ defines a Chern–Simons gauge theory on the Seifert manifold. By the CS/WZW correspondence [Witten, 1988], the CS theory on a manifold with boundary induces the $SU(3)_1$ WZW model on the boundary, whose Kac–Moody currents $J^a(z)$ ($a = 1, \dots, 8$) are the dynamical gluons (conformal weight $(1, 0) = \text{spin } 1$).

Part (a) is representation theory; part (b) is the Witten CS/WZW theorem [Witten, 1989].

Proof of part (a): representation structure. Five algebraic steps.

1. *Frobenius orbits (derived, not chosen).* The multiplicative group $(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/6\mathbb{Z}$ has exactly four subgroups. A non-abelian gauge symmetry requires a proper non-trivial subgroup Γ whose orbits form conjugate multiplets under CP (the palindromic involution $\sigma_6: m \mapsto 7-m$). We show $\Gamma = \mathbb{Z}/3\mathbb{Z}$ is the unique choice by exhaustive exclusion:

- (i) $\{1\}$: trivial, no gauge symmetry.
- (ii) $\mathbb{Z}/2\mathbb{Z} = \{1, 6\}$: acts by $m \mapsto -m$, the palindromic involution. This is conformal inversion $\xi \leftrightarrow 1/\xi$ on the Poincaré disk (Proposition 7.4), which reverses the orientation of the S^1 fiber in the Seifert geometry and is therefore parity P . Combined with the charge conjugation $C: m \mapsto N-m$ (equivalent on $\mathbb{Z}_N$ Fourier modes), this is the CP operator. It is a symmetry of the spectrum ($\lambda_m = \lambda_{N-m}$), not an internal gauge symmetry. Using it for gauge would conflate color with CP.
- (iii) $\mathbb{Z}/6\mathbb{Z}$: the full group. It produces a single orbit of size 6, which admits no decomposition into $\mathbf{3} \oplus \bar{\mathbf{3}}$.
- (iv) $\mathbb{Z}/3\mathbb{Z} = \{1, 2, 4\}$: the unique remaining subgroup. Orbits $O_+ = \{1, 2, 4\}$, $O_- = \{3, 5, 6\}$ are exchanged by CP ($\sigma_6(O_+) = O_-$), giving a genuine $\mathbf{3} \oplus \bar{\mathbf{3}}$.

The choice of generator (2 or $4 = 2^2$) is immaterial: the subgroup is unique, and both generators produce the same pair of orbits. Both orbits have the same Casimir multiset $\{3, 5, 6\}$: the gauge symmetry is exact at all orders of perturbation theory, not merely a linearized approximation.

2. *Cyclic action.* On O_+ : $\phi_1 \mapsto \phi_2 \mapsto \phi_4 \mapsto \phi_1$.

3. *McKay embedding.* Diagonalise: $\eta_k = \frac{1}{\sqrt{3}} \sum_j \omega^{jk} \phi_{2j}$ ($\omega = e^{2\pi i/3}$). Then $\sigma(\eta_1) = \omega \eta_1$, $\sigma(\eta_2) = \bar{\omega} \eta_2$, so $\sigma = \text{diag}(\omega, \omega^{-1}) \in SU(2)$: the McKay embedding $\mathbb{Z}/3 \hookrightarrow SU(2) \rightarrow A_2 = SU(3)$ [McKay, 1980, González-Sprinberg and Verdier, 1983].

4. *Klein quartic.* $X(7) = \Gamma(7) \backslash \mathbf{H}^2$ (genus 3, $\text{Aut} = \text{PSL}(2, \mathbb{F}_7)$, order 168). The differentials $H^0(X(7), \Omega^1)$ carry χ_3 ; σ generates $\mathbb{Z}/3 \subset \text{PSL}(2, \mathbb{F}_7)$.

5. *Eightfold Way.* $\chi_3 \otimes \bar{\chi}_3 = \chi_1 \oplus \chi_8$, $\text{Sym}^2(\chi_3) = \chi_6$, $\Lambda^2(\chi_3) = \bar{\chi}_3$. The adjoint χ_8 is irreducible under $\text{PSL}(2, \mathbb{F}_7)$. $\square$

Proof of part (b): gauge dynamics. Roadmap. Step B1 converts vortex displacements into CS connection perturbations (the dreibein chain). Step B2 extends this to all orders via the CS/WZW theorem. Steps B3–B4 apply the McKay correspondence and DHVW orbifold construction to *derive* the gauge group as $SU(3)$ at level 1 (see the derivation chain in §8.6 below). Step B5 promotes the boundary currents to 4D gauge bosons via KK reduction.

Step B1 (CS/WZW embedding). The charged Havelock modes η_1, η_2 (Step 3) are massive scalars on $\mathbf{H}^2$ with $m_k^2 = c_k(c_k - 1)$. Their boundary values parametrize $\mathbb{C}^2$, on

../figures/fig_frobenius_orbits.pdf

Figure 1: The Frobenius action $m \mapsto 2m \bmod 7$ on the seven Havelock modes. Blue: orbit $O_+ = \{1, 2, 4\}$ (quadratic residues). Red: orbit $O_- = \{3, 5, 6\}$ (non-residues). Grey: mode 0 (fixed by all automorphisms). The two orbits are exchanged by CP ($m \mapsto 7-m$), forming a $\mathbf{3} \oplus \mathbf{\bar{3}}$ conjugate pair.

which $\mathbb{Z}/3\mathbb{Z}$ acts as $\text{diag}(\omega, \omega^{-1}) \in \text{SU}(2)$. The CS/WZW theorem [Witten, 1988] gives the nonlinear identification: the boundary values of the CS gauge field A parametrize the WZW target manifold G , with the WZW action $S_{\text{WZW}}[g]$ equal to the on-shell CS action $S_{\text{CS}}[A|_{\partial}]$ for all field configurations, not just at the linearized level. The dreibein chain below verifies the linearized limit. A polygon vertex displacement δz_j perturbs the metric at the j -th cone point: $\delta g_{ab} = (\varepsilon_j/\pi) \partial_a \partial_b G(x, z_j) \delta z_j$. In the first-order formalism $g = e \otimes e$, this gives $\delta e^a = (1/2) e^b \delta g_{ab}$, and the CS perturbation is $\delta A^\pm = \delta \omega \pm \delta e/\ell$. The $\mathbb{Z}_N$ Fourier projection $\delta A_m = (1/N) \sum_j e^{-2\pi i j m/N} \delta A(z_j)$ decomposes into the mode- m CS perturbation, whose quadratic action is $|\delta A_m|^2 \lambda_m$, the Havelock eigenvalue. The charged modes η_1, η_2 (Step 3) are the $m \in O_+$ Fourier projections of δA . The $\mathbb{Z}/3\mathbb{Z}$ invariance is exact at all orders because it is a property of the Casimir spectrum, not of the coupling strength: the Frobenius $\sigma: m \mapsto 2m \bmod 7$ permutes modes within orbits $O_+ = \{1, 2, 4\}$ and $O_- = \{3, 5, 6\}$, whose Casimir multisets are both $\{3, 5, 6\}$.

Step B2 (nonlinear extension). The CS/WZW theorem (Witten 1988) extends the linearized identification of Step B1 to all orders: the boundary values of the CS gauge field parametrize the WZW target manifold G , with the WZW action equal to the on-shell CS action for all field configurations. The $\mathbb{Z}/3$ invariance of Step B1 therefore holds nonlinearly.

Step B3 (gauge group and level). The gauge group and its Chern–Simons level are determined by two results, each a theorem:

- *Gauge group:* $\text{SU}(3)$ (from McKay). The McKay correspondence for $\mathbb{Z}/3\mathbb{Z} \subset \text{SU}(2)$ gives the A_2 Dynkin diagram of $\text{SU}(3)$ [McKay, 1980, González-Sprinberg and Verdier, 1983].

- *Level:* $k = 1$ (from the ALE resolution and fusion ring). Two independent arguments fix $k = 1$: (a) *ALE/ K -matrix.* The orbifold singularity $\mathbb{C}^2/\mathbb{Z}_3$ has a minimal resolution whose exceptional divisors form a chain of (-2) -curves with intersection matrix equal to the A_2 Cartan matrix $K = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ [Kronheimer, 1989]. By the Wen K -matrix classification [Wen, 1995], the abelian CS theory with this K -matrix has the same topological order as the non-abelian $SU(3)_1$ CS theory: the discriminant group $\mathbb{Z}^2/K\mathbb{Z}^2 \cong \mathbb{Z}_3$ matches the three integrable representations of $SU(3)_1$, and the mutual statistics encoded in $K^{-1} \bmod 1$ match the $SU(3)_1$ braiding data. (b) *Fusion ring.* The representation ring of $\mathbb{Z}/3 \subset SU(2)$ is isomorphic to the fusion ring of $SU(3)_k$ only at $k = 1$ [Cappelli et al., 1987]: at $k = 1$ there are exactly three integrable representations (trivial, fundamental, anti-fundamental), matching the three irreducible representations of $\mathbb{Z}/3$; at $k \geq 2$ the fusion ring has additional representations with no $\mathbb{Z}/3$ counterpart. This singles out $k = 1$ uniquely.

Why $k = 1$ is exact. The homotopy group $\pi_3(SU(3)) = \mathbb{Z}$ forces the CS level to be a positive integer. The unique crepant (minimal) resolution of $\mathbb{C}^2/\mathbb{Z}_3$ gives the intersection form $K = A_2$ without rescaling; a non-minimal resolution would introduce $(-n)$ -curves with $n > 2$, changing K , and a multi-cover would rescale $K \rightarrow kK$ [Kronheimer, 1989]. Thus $k = 1$ is the unique level compatible with the crepant resolution. This is a topological invariant, independent of the boundary central charge c or any continuous parameter of the theory. The gauge sector ($SU(3)_1$ CS) and the gravitational sector ($SL(2, \mathbb{R})$ CS at level $k_{\text{grav}} = 2b(N)$) are independent theories on the same manifold, with independent levels: $k_{\text{gauge}} = 1$ is compact and integer; $k_{\text{grav}} \approx 8.6$ is non-compact and irrational.

Remark 8.2 (Boundary and DHVW consistency checks). *Two independent calculations confirm $k = 1$:* (i) *The DHVW orbifold construction [Dixon et al., 1985] on the boundary CFT at $c = 12b(N)$ yields a twisted-sector conformal dimension $h_k = k(3-k)/9 = 2/9$, which receives a Liouville dressing correction $\delta h = O(h_k^2/c) \approx 10^{-3}$ on the $\mathbf{H}^2$ background. The boundary level identification gives $k = 1 + O(10^{-3})$, consistent with the exact topological result. The orbifold at irrational c is a consistent (non-rational) CFT: unitarity holds at $c > 1$ by the Friedan–Qiu–Shenker theorem; modular invariance of $Z_{\text{orb}} = (1/|G|) \sum_{g,h} Z_g^h$ follows from the geometric transformation rules at every c [Vafa, 1986]; OPE closure follows because $\mathbb{Z}/3$ is an inner automorphism.* (ii) *The representation ring of $\mathbb{Z}/3 \subset SU(2)$ is isomorphic to the fusion ring of $SU(3)_1$ [Cappelli et al., 1987], consistent with $k = 1$ and no other level.*

Step B4 (G_L invariance). The Frobenius $\mathbb{Z}/3\mathbb{Z}$ acts as an exact automorphism of the full nonlinear boundary CFT. The proof uses the $G_L \times G_R$ invariance of the WZW action: left multiplication $g(x) \mapsto h \cdot g(x)$ by $h \in \mathbb{Z}/3\mathbb{Z} \subset G$ preserves both the kinetic term $\text{Tr}(g^{-1}\partial g \cdot g^{-1}\bar{\partial} g)$ and the Wess–Zumino term $\int \text{Tr}(g^{-1}dg)^3$ (both are manifestly G_L -invariant). This gives exact invariance of the action, the path integral, all n -point functions (Schwinger–Dyson), and all OPE coefficients. The Casimir multiset invariance follows: the Havelock eigenvalues are spectral data of the G_L -invariant theory. *Step B5 (CS/WZW correspondence $\rightarrow$ 4D gauge field).* The chain from orbifold current algebra to 4D gauge theory has three links, each a theorem:

1. $\widehat{\mathfrak{su}}(3)_1$ current algebra on the boundary $\rightarrow SU(3)_1$ Chern–Simons theory in the bulk (Witten CS/WZW theorem [Witten, 1988]);
2. CS_3 on $\mathbf{H}^2 \times S^1 \rightarrow YM_2$ on $\mathbf{H}^2$ (KK reduction: the CS action $\int \text{Tr}(A \wedge dA + \frac{2}{3}A^3)$ on $\mathbf{H}^2 \times S^1$, after Fourier expanding $A = A^{(0)} + \sum_{n \neq 0} A^{(n)} e^{in\theta}$ and integrating over S^1 ,

yields the 2D YM action $\int_{\mathbf{H}^2} \text{Tr}(F^2)$ for the zero mode $A^{(0)}$, with coupling $g^2 = 4\pi/k$ and massive KK modes $M_n = n/R_{S^1}$;

3. YM₂ on $\mathbf{H}^2$ + WDW time direction $\rightarrow$ YM₃₊₁: the breathing mode $\rho(t)$ (the scale factor of the $\mathbf{H}^2$ slice) satisfies the WDW equation $H\Psi = 0$ (Paper III, Proposition 3.8). The Feynman–Kac theorem identifies the WDW evolution parameter with Euclidean time (Paper III, §13): the WKB solution $\Psi_0 \propto e^{-\int V d\rho}$ evolves monotonically in ρ , so ρ is a good time coordinate. The gauge field $A^{(0)}$ on the spatial $\mathbf{H}^2$ thereby acquires the time direction ρ , giving YM₂₊₁. The KK tower $A^{(n)}$ with $n \geq 1$ provides the additional spatial dimension (the S^1 fiber), completing $1 + 2 + 1 = 4$ dimensions.

The SU(3) gauge group is therefore *determined* by the Frobenius action on the Havelock modes (via McKay + DHVW) from the same inputs (H1)–(H3). In the topological bulk, there are zero propagating bulk modes; the boundary currents $J^a(z)$ ($a = 1, \dots, 8$, conformal weight (1, 0), spin 1) are the dynamical degrees of freedom. After KK reduction on S^1 , the boundary currents become 4D spin-1 gauge bosons with the SU(3) vertex structure inherited from the Kac–Moody OPE, at central charge

$$c_{\text{SU}(3)} = \frac{k \dim(\text{SU}(3))}{k+h^\vee} = \frac{8}{4} = 2. \quad (8.5)$$

The total boundary central charge is $c_{\text{tot}} = 12b(N)$. The gauge contributions ($c_{\text{SU}(3)} = 2$, $c_{\text{SU}(2)} = 1$, $c_{\text{U}(1)} = 1$) are subsumed: $c_{\text{grav}} = 12b(N) - 4$, with $G = 3/(2c_{\text{tot}}) = 3/(2 \times 12b(N))$ using the total central charge in the Brown–Henneaux relation. $\square$

8.4 Strong CP problem: $\theta_{\text{QCD}} = 0$ from level quantization

The QCD vacuum angle is $\theta_{\text{QCD}} = 0$ exactly, by the following three-step argument.

1. *CS level quantization.* For a compact gauge group, $\pi_3(\text{SU}(3)) = \mathbb{Z}$ forces the CS level $k \in \mathbb{Z}^+$. At $k = 1$: the vacuum angle is $\theta = 2\pi k_{\text{frac}} = 0$ (integer level, no fractional part).
2. *Fermion shifts preserve integrality.* Each massive Dirac fermion in representation R shifts the CS level by $\Delta k = T(R) \text{sign}(m)$ (parity anomaly; Redlich 1984), where $T(R)$ is the Dynkin index ($T(\text{fund}) = 1/2$ for SU(3)). For 6 quark flavors in the fundamental: $\Delta k = 6 \times T(\text{fund}) \times \text{sign}(m) = 6 \times 1/2 = 3 \in \mathbb{Z}$ (the three colors enter through T , not as independent fermions). All masses have the same sign (from a single Higgs VEV), giving $k_{\text{eff}} = 1 + 3 = 4$ (integer).
3. *Gravitational sector decouples.* The gravitational CS level $k_{\text{grav}} = 2b(N) = 8.60$ is irrational, with fractional part $k_{\text{frac}}^{\text{grav}} = 0.60$. This does not affect θ_{QCD} because the gauge and gravitational traces are orthogonal: $\text{Tr}(T_a^{\text{SU}(3)} T_b^{\text{SL}(2, \mathbb{R})}) = 0$.

At all loop orders, $k_{\text{eff}}^{\text{SU}(3)} \in \mathbb{Z}$ and $\theta_{\text{QCD}} = 0$. The mechanism is topological (level quantization from π_3), requires no axion, and predicts: vanishing neutron EDM at all orders, null results at ADMX and ABRACADABRA (no axion field, no axion dark matter).

Orbit	Modes	Casimirs f	σ -eigenvalue
O_+	$\{1, 2, 4\}$	$\{3, 5, 6\}$	$1, \omega, \omega^2$
O_-	$\{3, 5, 6\}$	$\{6, 5, 3\}$	$1, \omega, \omega^2$

The QR/QNR orbits $O_\pm$ are the $\chi_3/\bar{\chi}_3$ sectors of Theorem 8.5: modes $\{1, 2, 4\}$ (quadratic residues mod 7) carry χ_3 , modes $\{3, 5, 6\}$ (non-residues) carry $\bar{\chi}_3$.

Consistency check ($C_F = 1 + 2B_2$). The fundamental Casimir $C_F = (N_c^2 - 1)/(2N_c) = 4/3$ at $N_c = 3$ equals $1 + 2B_2$, where $2B_2 = 1/3$ is the Laurent constant of the csc^2 kernel (Paper I, §7.7). $C_F = 1 + 2B_2$ holds *only* at $N_c = 3$ (the unique positive root of $3N_c^2 - 8N_c - 3 = 0$), which is the value determined by the Frobenius order $\text{ord}_7(2) = 3$. The identity $2B_2 = 1/N_c$ connects the csc^2 self-interaction constant to the gauge group rank: the *same kernel* that determines the self-interaction (B_2) also determines the gauge group (N_c) via the Frobenius orbits. This is a consistency check between two independently derived quantities, not a derivation of either.

8.5 U(1) from Kaluza–Klein

The S^1 fiber with flux $N/2$ gives a U(1) gauge field at all N , with coupling $\alpha = 6/(\pi c)$ and the exact lock $\alpha/(8\pi G) = 1/(2\pi^2)$. This is standard Kaluza–Klein reduction. The KK U(1) is the *only* abelian gauge symmetry in the theory: every KK mode carries charge $Q = m/N$ fixed by its S^1 momentum quantum number. This U(1) *is* hypercharge: it is the only abelian gauge symmetry in the theory, and the charge normalization $Q = m^*/N = 1/2$ at the critical $N = 4$ mode is the Standard Model hypercharge of the Higgs doublet, derived from the KK spectrum.

8.6 Summary: the derivation chain

The gauge group $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ at levels $k = 1$ is *derived* from the Seifert geometry $\mathbf{H}^2 \times_N S^1$. The chain uses three mathematical theorems applied to the geometry:

1. *SU(2) from CS gravity* (Witten 1988): KK reduction on S^1 gives 2+1D gravity = CS[SU(2)]. At $N = 4$, the critical Havelock eigenvalue $f(2, 4) = 2$. The Euler class $e = N/2$ breaks $\text{SU}(2)_R$ via the parity anomaly.
2. *SU(3) from McKay* (McKay 1980): The unique $\mathbb{Z}/3\mathbb{Z}$ subgroup of $(\mathbb{Z}/7\mathbb{Z})^*$ (selected by CP exclusion and conjugate-pair requirement; see Step 1 of Theorem 8.5) gives orbits $\{1, 2, 4\}$, $\{3, 5, 6\}$. The McKay correspondence maps $\mathbb{Z}/3 \subset \text{SU}(2) \rightarrow A_2 \rightarrow \text{SU}(3)$. The DHVW twist-field OPE (1985) fixes the level $k = 1$ (up to corrections $O(h^2/c) \sim 0.1\%$, which are small enough that integer quantization of the CS level forces $k = 1$ exactly).
3. *U(1) from KK*: The fiber gives a U(1) gauge field with $k_Y = 1$ (single-valued holonomy). Charge $Q = m/N$ is fixed by the S^1 momentum quantum number.

The CS-Havelock identity (Paper I, Theorem A.2) proves that the Casimir $f(m, N) = m(N - m)/2$ is simultaneously the vortex eigenvalue, the $\mathfrak{sl}(2, \mathbb{R})$ Casimir (Helgason), and the CS Wilson line Casimir (Etingof–Frenkel–Kirillov). These are the *same operator* (the Laplacian on $\mathbf{H}^2$) restricted to the same subspace ($\mathbb{Z}_N$ tangential modes).

Central charge budget (no-hidden-sector theorem): An exhaustive enumeration of simple Lie groups G at level $k \leq 3$ yields 33 combinations with total WZW central charge $c_{\text{gauge}} = 4$.³ Of these, $\text{SU}(3)_1 \times \text{SU}(2)_1 \times \text{U}(1)_1$ is the unique one containing all three derived factors. Since $c_{\text{gauge}} = 2 + 1 + 1 = 4$ is fully accounted for, no additional gauge sector can fit within the central charge budget: the gauge group is *exactly* $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$, with no hidden factors.

Remark 8.3 (Derivation chain from the two axioms). *Given $N = 7$ (A1) and the Seifert geometry (A2), the derivation chain uses five mathematical theorems (Witten, EFK, Havelock, McKay, DHVW) and the OPS entropy principle. The $\mathbb{Z}/3\mathbb{Z}$ orbifold group is determined by the Frobenius structure: it is the unique proper non-trivial subgroup of $(\mathbb{Z}/7\mathbb{Z})^*$ not already identified with CP (Step 1 of Theorem 8.5). The assignment of KK pairs to fermion generations and the identification of pair 3 with the Higgs doublet follow from the BF-crossing structure (only pair 3 has $c_{\text{up}} < 1 < c_{\text{dn}}$). The CKM phase $\delta = 69.3^\circ$ is a mathematical computation on $\mathbf{H}^2$; its physical interpretation as the CP-violating phase relies on the vortex-gravity isomorphism of Paper III.*

9 The Weinberg angle

Derivation 9.1 (Weinberg angle from polygon eigenvalues). *The KK reduction of CS_3 on $\mathbf{H}^2 \times S^1$ gives YM_2 on $\mathbf{H}^2$ with coupling $g^2 = 2\pi/((k+h^\vee)R)$ (Blau–Thompson 1993). The effective coupling of matter in the m -th Fourier mode is $\alpha(m) = g^2 \times f(m, N)$, where $f(m, N) = m(N-m)/2$ is the Havelock eigenvalue, the eigenvalue of the $\mathbf{H}^2$ Laplacian restricted to the $\mathbb{Z}_N$ mode (Proposition 2.1). This is a geometric quantity (sector G of Theorem 9.2), determined by the polygon, not by a gauge-group representation label. The common factor $2\pi/R$ cancels in the mixing angle ratio:*

$$\sin^2 \theta_W = \frac{h_Y}{h_Y + h_W} = \frac{Q^2(k+h^\vee)}{Q^2(k+h^\vee) + k_Y f(m^*, N)} = \frac{3}{11} \approx 0.2727, \quad (9.1)$$

where $h_W = f(m^*, N)/(k+h^\vee) = 2/3$ (Havelock eigenvalue at $N = 4$, $m^* = 2$), $h_Y = Q^2/k_Y = 1/4$ (KK momentum charge $Q = m^*/N = 1/2$), and $k_Y = 1$ ($\text{U}(1)$ compact-boson level). No representation label j is needed: the number 2 enters as $f(2, 4)$, an eigenvalue of the $\mathbf{H}^2$ Laplacian proved in Paper I. The integrability bound $j \leq k/2$ of $\text{SU}(2)_1$ WZW constrains topological states in the CS Hilbert space; it does not constrain geometric eigenvalues on $\mathbf{H}^2$.

Proof. Three steps: derive the quantum numbers from the Casimir, derive the CS gauge couplings from the DHVW levels, compute the mixing angle from the KK-reduced action.

Step 1 (Polygon eigenvalue = gauge Casimir). The critical mode at $N = 4$ has Havelock eigenvalue $f(2, 4) = m^*(N-m^*)/2 = 2$ and KK charge $Q = m^*/N = 1/2$. Both are read off from the vortex spectrum (Paper I); no choice is involved. The CMS–CS Casimir identification (Proposition 2.1) proves that $f(m, N)$ is simultaneously the $\mathbf{H}^2$ Laplacian eigenvalue (geometric), the $\mathfrak{sl}(2, \mathbb{R})$ Casimir (representation-theoretic), and the CS Wilson line eigenvalue (gauge-theoretic), all equal to $m(N-m)/2$ because all three use the same operator on the same space (Helgason symmetric space theorem). The value $f(2, 4) = 2$ is therefore not a “choice of representation”: it is the unique eigenvalue

³The 33 combinations include $\text{SU}(5)_1$, $\text{SO}(8)_1$, $\text{SU}(3)_3$, and 30 multi-factor products; the full list is in the companion code (`gauge_group_derivation.py`).

of the $m = 2$ mode on $\mathbf{H}^2$, which the CMS–CS identification proves equals the $\text{SU}(2)$ Casimir $C_2 = j(j+1) = 2$ at $j = 1$. The fundamental representation ($j = 1/2$, $C_2 = 3/4$) does not enter because the critical Fourier mode at $N = 4$ has $m^* = 2$, not $m^* = 1$; the polygon determines which mode is critical, and the Casimir identification determines its gauge-theoretic value.

Step 2 (CS gauge couplings). At the Chern–Simons threshold [Witten, 1988]:

- $\text{SU}(2)$: $g_W^2 = 2\pi/((k+h^\vee)R) = 2\pi/(3R)$ ($k = 1$, $h^\vee(\text{SU}(2)) = 2$);
- $\text{U}(1)$: $g_Y^2 = 2\pi/(k_Y R) = 2\pi/R$ ($k_Y = 1$).

The level $k = k_Y = 1$ is forced at each gauge group by the ALE resolution (Theorem 8.5, Step B3):

- $\text{SU}(3)$: the Frobenius at $N = 7$ generates $\mathbb{Z}/3\mathbb{Z}$; $k = 1$ by the ALE/ K -matrix and fusion ring arguments of Step B3.
- $\text{SU}(2)$: the Frobenius at $N = 4$ generates $\mathbb{Z}/2\mathbb{Z}$; the ALE resolution of $\mathbb{C}^2/\mathbb{Z}_2$ has A_1 intersection form $K = [2]$, and the fusion ring of $\mathbb{Z}/2$ matches $\text{SU}(2)_1$ uniquely (two integrable representations: trivial and fundamental).
- $\text{U}(1)$: $k_Y = 1$ from single-valued holonomy on S^1 .

In all three cases, $k = 1$ is a topological invariant of the resolved singularity. As a cross-check: $k = 2$ would give $\sin^2 \theta_W = 1/3$, and $k = 3$ gives $5/13$; both are experimentally excluded.

Step 3 (Mixing angle from KK-reduced Yang–Mills couplings). The KK reduction of CS_3 on $\mathbf{H}^2 \times S^1(R)$ at level k gives YM_2 on $\mathbf{H}^2$ with bare coupling $g^2 = 2\pi/((k+h^\vee)R)$ (Blau and Thompson 1993; the shift $k \rightarrow k+h^\vee$ is the standard one-loop CS renormalization). The gauge–matter vertex couples with strength $g \times T_R^a$ (the Lie-algebra generator in representation R), so the *effective* coupling for matter in representation R is

$$\alpha_{\text{eff}}(R) = g^2 \times C_2(R) = \frac{2\pi C_2(R)}{(k+h^\vee)R}. \quad (9.2)$$

The Lie-algebra Casimir $C_2(R) = \sum_a (T_R^a)^2$ is a property of the representation, independent of k . The fiber radius R cancels in the mixing angle ratio:

$$\sin^2 \theta_W = \frac{g_Y^2 C_2(Y)}{g_Y^2 C_2(Y) + g_W^2 C_2(W)} = \frac{Q^2/k_Y}{Q^2/k_Y + f(m^*, N)/(k+h^\vee)}. \quad (9.3)$$

The numerator and denominator are products of two independent factors: the YM coupling $g^2 = 2\pi/((k+h^\vee)R)$ (determined by the CS level via KK reduction) and the Lie-algebra Casimir C_2 (a property of the representation, independent of k). Substituting from Steps 1–2 ($Q^2 = 1/4$, $k_Y = 1$, $f(2, 4) = 2$, $k+h^\vee = 3$):

$$\sin^2 \theta_W = \frac{1/4}{1/4 + 2/3} = \frac{3}{11}.$$

□

Theorem 9.2 (Gauge–matter factorization). *The theory has three cleanly separated sectors:*

- (T) Topological sector (*3D CS*): determines the gauge group and level k . The WZW integrability bound $j \leq k/2$ constrains this sector (conformal blocks, Verlinde formula, anyonic braiding).
- (D) Dynamical sector (*2D KK-reduced YM*): the KK reduction of CS_3 on $\mathbf{H}^2 \times S^1(R)$ produces YM_2 on $\mathbf{H}^2$ with coupling $g^2 = 2\pi/((k+h^\vee)R)$ (Blau–Thompson).
- (G) Geometric sector (*polygon on $\mathbf{H}^2$*): the Havelock critical mode has Casimir $C_2 = f(m^*, N) = m^*(N-m^*)/2$, an eigenvalue of the $\mathbf{H}^2$ Laplacian determined by the polygon alone.

The effective coupling factorizes as

$$\alpha(R) = \underbrace{g^2}_{(D) \text{ gauge}} \times \underbrace{C_2(R)}_{(G) \text{ matter}}, \quad (9.4)$$

where g^2 comes from sector (D) and C_2 from sector (G). The integrability bound from sector (T) is irrelevant: it constrains topological excitations (Wilson loops), not geometric excitations (polygon displacements).

Proof. (a) From CS to YM (sectors $T \rightarrow D$). The CS action $S_{\text{CS}} = (k/4\pi) \int \text{Tr}(A \wedge dA + \frac{2}{3}A^3)$ on $\mathbf{H}^2 \times S^1(R)$ has no propagating degrees of freedom in 3D. The KK reduction on $S^1(R)$ produces a 2D YM kinetic term $\frac{1}{4g^2} \int F^2$ with coupling $g^2 = 2\pi/((k+h^\vee)R)$ (Blau and Thompson 1993; the shift $k \rightarrow k+h^\vee$ is the standard one-loop CS renormalization, which depends only on the root system A_1 and takes the same value $h^\vee = 2$ for both the compact form $\text{SU}(2)$ and the non-compact form $\text{SL}(2, \mathbb{R})$). This is the step that creates a propagating gauge field.

(b) *Geometric matter* (sector G). The vortex critical mode at $N = 4$ has Havelock eigenvalue $f(2, 4) = 2 \times 2/2 = 2$. This is a *geometric* quantity: the eigenvalue of the $\mathbf{H}^2$ Laplacian restricted to the $m = 2$ Fourier mode. The mode exists independently of the gauge field, it is a polygon displacement on $\mathbf{H}^2$, present before the CS theory is introduced. It couples to the gauge field through the standard covariant derivative $D_\mu = \partial_\mu - ig A_\mu^a T_R^a$ of the KK-reduced YM theory, giving $\alpha = g^2 C_2(R)$ at one-gluon exchange.

(c) *Why the integrability bound does not apply.* The WZW integrability bound $j \leq k/2$ constrains the *compact* $\text{SU}(2)$ fusion rules: the quantum dimension $d_j = \sin((2j+1)\pi/(k+2))/\sin(\pi/(k+2))$ vanishes at $k = 1, j = 1$ ($d_1 = 0$), so the $j = 1$ representation does not exist as a topological (anyonic) excitation. The Casimir $C_2 = f(2, 4) = 2$, however, is not a compact $\text{SU}(2)$ quantum number. It is the eigenvalue of the $\mathbf{H}^2$ Laplacian for the $m = 2$ Fourier mode (Paper I), which the CMS–CS identification (Proposition 2.1) proves equals the $\mathfrak{sl}(2, \mathbb{R})$ Casimir in the discrete series at $j = 1$. The non-compact group $\text{SL}(2, \mathbb{R})$ has *no* integrability bound: its discrete series contains representations at all $j \geq 1/2$, independent of any CS level. The distinction between the two groups is physical:

- *Compact* $\text{SU}(2)$ (sector T): governs topological braiding of anyons. At $k = 1$, only $j = 0$ and $j = 1/2$ anyons exist. The vanishing $d_1 = 0$ means $j = 1$ Wilson loops decouple from the topological sector.
- *Non-compact* $\text{SL}(2, \mathbb{R})$ (sector G): governs the geometry of $\mathbf{H}^2$. The KK mode $m = 2$ has Casimir $j(j+1) = 2$ in the discrete series. No truncation applies.

(c') *Operator identity: the gauge Casimir IS the Havelock eigenvalue.* The groups $SU(2)$ and $SL(2, \mathbb{R})$ are not independent here. These are *real forms of the same complex Lie algebra* $\mathfrak{sl}(2, \mathbb{C})$, and the Casimir operator $\Omega = \sum_a T^a T^a$ is defined over $\mathfrak{sl}(2, \mathbb{C})$ with the *same* eigenvalues $j(j+1)$ regardless of real form. The proof that $C_2 = f(m^*, N)$ in the gauge coupling uses no identification across groups:

1. *Operator identity* (Witten + Helgason). The CS connection on $\mathbf{H}^2$ is $A = \omega + e/\ell$ (spin connection + dreibein; [Witten 1988](#)). The covariant Laplacian built from this connection, $D^2 = (\partial_\mu - ig A_\mu^a T^a)^2$, equals the $\mathbf{H}^2$ Laplace–Beltrami operator $\Delta_{\mathbf{H}^2}$, because A is the gravitational connection on $\mathbf{H}^2 = SL(2, \mathbb{R})/SO(2)$. By the Helgason symmetric space theorem (?), $\Delta_{\mathbf{H}^2} = \Omega_{\mathfrak{sl}(2, \mathbb{R})}$ on $SL(2, \mathbb{R})$ -equivariant functions. Therefore:

$$D_{\text{CS}}^2 = \Delta_{\mathbf{H}^2} = \Omega_{\mathfrak{sl}(2, \mathbb{R})}.$$

(The Casimir of $\mathfrak{sl}(2, \mathbb{R})$ extends to $\mathfrak{sl}(2, \mathbb{C})$ by complexification; the eigenvalue $j(j+1)$ is preserved because both real forms share the same Killing form normalization.) All three are the *same* operator on the *same* Hilbert space (acting on scalar-type KK modes, which are $\mathbb{Z}_N$ -equivariant functions on $\mathbf{H}^2 = SL(2, \mathbb{R})/SO(2)$).

2. *Eigenvalue extraction.* The KK mode ϕ_{m^*} is an eigenfunction: $D^2 \phi_{m^*} = f(m^*, N) \phi_{m^*}$, where $f(m^*, N) = m^*(N-m^*)/2$ is the Havelock Casimir (Paper I). At $m^* = 2$, $N = 4$: $f(2, 4) = 2$.
3. *Representation content.* On the symmetric space $\mathbf{H}^2 = SL(2, \mathbb{R})/SO(2)$, the Plancherel decomposition of $L^2(\mathbf{H}^2)$ into irreducible representations of $SL(2, \mathbb{R})$ is multiplicity-free (? Ch. III, Thm. 1.3): each eigenspace $E_\lambda = \{f : \Delta f = \lambda f\}$ is a single irreducible representation. The eigenvalue $\lambda = j(j+1)$ selects the discrete series D_j^+ ; there is no other representation with the same eigenvalue. At $\lambda = 2$: $j = 1$, so $\phi_{m^*} \in D_1^+$ (the three-dimensional discrete series representation). Under Euclidean continuation $SL(2, \mathbb{R}) \rightarrow SU(2)$, D_1^+ maps to the adjoint (spin-1, dimension 3). The mode ϕ_{m^*} therefore transforms as a *triplet* under $SU(2)$. This is confirmed by the dreibein chain (Step B1): δA_{m^*} is a connection perturbation, and connection perturbations transform in the adjoint by definition ($\delta A \rightarrow g \delta A g^{-1} + g dg^{-1}$).
4. *Gauge coupling extraction.* The KK-reduced YM Lagrangian couples matter to the gauge field with vertex strength $g T_R^a$, giving the effective coupling $\alpha = g^2 C_2(R)$, where $C_2(R) = \sum_a (T_R^a)^2$ is the quadratic Casimir of the matter representation R . By Steps 1–3, R is the adjoint ($j = 1$, D_1^+ of $SL(2, \mathbb{R})$) and $C_2 = j(j+1) = f(m^*, N) = 2$. The eigenvalue determines the representation (multiplicity-free, Step 3) and the representation determines the Casimir.
5. *Result.* $\alpha_{\text{eff}} = g^2 \times C_2 = g^2 \times f(m^*, N) = [2\pi/((k+h^\vee)R)] \times m^*(N-m^*)/2$. The gauge coupling $g^2 = 2\pi/(3R)$ (from CS level $k = 1$, $h^\vee = 2$; Blau–Thompson KK reduction); the Casimir $C_2 = 2$ (from the $\mathbf{H}^2$ Laplacian eigenvalue, Step 2). No step invokes a representation label, an integrability bound, or an identification across different groups.

(d) *The Weinberg angle uses only sectors D and G.* The effective couplings are $\alpha_W = g_W^2 \cdot C_2(j=1) = (2\pi/3R) \cdot 2$ and $\alpha_Y = g_Y^2 \cdot Q^2 = (2\pi/R) \cdot 1/4$. The common factor $2\pi/R$ cancels in the mixing ratio: $\sin^2 \theta_W = \alpha_Y/(\alpha_Y + \alpha_W) = 3/11$. The topological sector (where $d_1 = 0$) does not enter.

(e) *Cross-check:* $SU(3)_1$. Gluons have $C_2(\text{adj}) = 3$, also non-integrable at $k = 1$. They propagate through the YM kinetic term (sector D), not through topological braiding (sector T). This is standard in all CS–matter theories [?]: the CS level determines the gauge coupling, the matter representation is a separate physical input. $\square$

Comparison with unification frameworks.

Framework	Definition at threshold	Value
SM alone	$g'^2/(g^2 + g'^2)$	free parameter
$SU(5)$ GUT	$\text{Tr}(Y^2)/\text{Tr}(T_3^2 + Y^2)$	$3/8 = 0.375$
CS/WZW (this work)	$h_Y/(h_Y + h_W)$	$3/11 = 0.273$

In 4D YM, the gauge coupling g is representation-independent, so $\sin^2 \theta_W = g'^2/(g^2 + g'^2)$ requires an external normalization (the GUT embedding). In the KK-reduced framework, the $\mathbf{H}^2$ Laplacian eigenvalue $f(m, N)$ enters the vertex coupling $\alpha = g^2 f(m, N)$ (eq. (9.4)), making the coupling representation-dependent at the CS threshold. The mixing angle $h_Y/(h_Y + h_W)$ is then the ratio of *conformal weights*, which are directly measurable through scattering cross-sections. No GUT embedding is needed because the polygon (sector G) and the CS level (sector D) together fix the ratio.

At the CS threshold there is a single matter species (the Havelock critical mode at $N = 4$). Below the polygon scale the theory flows to the 4D SM, and $3/11$ becomes the boundary condition for the standard RG running (one-loop, $b_1 = 41/10$, $b_2 = -19/6$), which brings it to $\sin^2 \theta_W(M_Z) = 0.231$, matching the measured value 0.2312 to 0.1% . This determines $M_{\text{poly}} \approx 300$ TeV: a consistency check (the running must land in the 50–500 TeV window). The $SU(5)$ prediction $3/8$ matches at $M_{\text{GUT}} \sim 10^{14}$ GeV (62% above experiment at tree level).

9.1 Spectral derivation of the electroweak couplings

The CS inverse couplings that enter the Weinberg angle can be derived independently from the spectral structure of the $\tilde{E}_8$ Cartan matrix (Paper I, §9; Paper V, §??). This provides a graph-theoretic derivation of the coupling constants that enter (9.3), without invoking Chern–Simons theory directly.

Lemma 9.3 (Spectral–CS bridge). *The spectral denominators $\{1, 2, 3, 5\}$ of the $\tilde{E}_8$ Cartan matrix are the stabilizer orders of the icosahedral symmetry axes (trivial, edge $\mathbb{Z}_2$, face $\mathbb{Z}_3$, vertex $\mathbb{Z}_5$). The dual Coxeter numbers of the SM gauge groups equal the corresponding stabilizer orders: $h^\vee(SU(3)) = 3 = |\mathbb{Z}_3|$ (face), $h^\vee(SU(2)) = 2 = |\mathbb{Z}_2|$ (edge), $h^\vee(U(1)) = 0$ (trivial). Consequently, the CS inverse couplings at level $k = 1$ are*

$$1/g^2(SU(3)) = 1 + 3 = 4, \quad 1/g^2(SU(2)) = 1 + 2 = 3, \quad k_Y = 1. \quad (9.5)$$

The vertex stabilizer $\mathbb{Z}_5$ gives $h^\vee(SU(5)) = 5$ (the GUT group), which breaks to $SU(3) \times SU(2) \times U(1)$ at the $K = 0$ transition.

Proof. The spectral denominators $k = h/\text{gcd}(m, h)$ for the affine m -values are: $k = 1$ (boundary, $m \in \{0, 30\}$), $k = 2$ (Majorana, $m = 15$), $k = 3$ (triangle, $m \in \{10, 20\}$), $k = 5$ (golden, $m \in \{6, 12, 18, 24\}$).

Each prime $p \mid h = 30$ corresponds to an icosahedral rotation axis with stabilizer $\mathbb{Z}_p$: $p = 5$ (vertices, 12 elements of order 5 or 10), $p = 3$ (faces, 20 elements of order 3 or 6),

$p = 2$ (edges, 30 elements of order 4). The orbifold gauge group at each axis is $SU(p)$ via the McKay correspondence $\mathbb{Z}_p \subset SU(2) \rightarrow \tilde{A}_{p-1}$, with $h^\vee(SU(p)) = p$.

For the SM gauge groups after breaking: $SU(3)$ retains $h^\vee = 3$ from the face stabilizer, $SU(2)$ retains $h^\vee = 2$ from the edge stabilizer, and $U(1)$ is abelian ($h^\vee = 0$). The universal CS level $k = 1$ (Hopf Euler class, §9) gives (9.5). $\square$

Corollary 9.4 (Spectral Weinberg angle). *The spectral denominators of the electroweak sectors recover*

$$\sin^2 \theta_W = \frac{Q^2/k_Y}{Q^2/k_Y + f(m^*, N)/(k+h^\vee)} = \frac{1/4}{1/4 + 2/3} = \frac{3}{11} = \frac{a-1}{a+N_{\text{crit}}}, \quad (9.6)$$

where $a = k_1(E_8) = 4$ and $N_{\text{crit}} = 7$. The formula factorizes into two independent contributions:

- Spectral (from the $\tilde{E}_8$ graph): $k_W/k_Y = 3$ (the CS coupling ratio from Lemma 9.3).
- Havelock (from the vortex spectrum on $\mathbf{H}^2$): $C_2/Q^2 = 2/(1/4) = 8$ (the Casimir-to-charge ratio from the critical mode $m^* = 2$, $N = 4$).

The Weinberg angle is $\sin^2 \theta_W = 3/(3+8) = 3/11$: the spectral ratio divided by the sum of spectral and Havelock ratios.

Proof. By the spectral–CS bridge (Lemma 9.3): $k + h^\vee(SU(2)) = 3 = k_W$ and $k_Y = 1$. By the Havelock Casimir (Step 1, §9): $j = 1$ ($C_2 = 2$) and $Q = 1/2$ ($Q^2 = 1/4$), where $N = 4$ is the unique polygon in $[3, N_{\text{crit}}]$ whose critical Casimir $f(\lfloor N/2 \rfloor, N)$ gives a positive integer $j < 2$ (gauge boson, not graviton; see Step 1 for the exhaustive check). Then $\alpha_Y/\alpha_W = (Q^2 \cdot k_W)/(C_2 \cdot k_Y) = (1/4 \cdot 3)/(2 \cdot 1) = 3/8$, and $\sin^2 \theta_W = (3/8)/(1+3/8) = 3/11$. The numerator $3 = k_W/k_Y$ is the spectral contribution; the denominator $11 = 3+8$ adds the Havelock contribution $C_2 \cdot k_Y/Q^2 = 8$. $\square$

Lemma 9.5 (Bernoulli–Coxeter identity (classical)). $B_4 = B_8 = -1/30 = -1/h(E_8)$. Consequently $\zeta(-3) = 1/120 = 1/|I^*|$ and $\zeta(-7) = 1/240 = 1/|\text{roots}(E_8)|$.

Proof. By von Staudt–Clausen, $\text{denom}(B_{2k}) = \prod_{(p-1)|2k} p$. For B_4 : $(p-1)|4$ gives $p \in \{2, 3, 5\}$, $\text{denom} = 30$. For B_8 : $(p-1)|8$ gives $p \in \{2, 3, 5\}$ (since $p = 9 = 3^2$ is not prime), $\text{denom} = 30$. Both numerators are -1 by the Bernoulli recursion. Then $\zeta(-3) = -B_4/4 = (1/30)/4 = 1/120$ and $\zeta(-7) = -B_8/8 = (1/30)/8 = 1/240$. $\square$

Remark 9.1. The identity $B_4 = B_8$ relies on $\text{rank}(E_8) + 1 = 9 = 3^2$ being composite. If 9 were prime, $\text{denom}(B_8) = 270 \neq 30$, and the Riemann zeta would not encode $|I^*|$ at $s = -3$. The coincidence $B_4 = B_8$ depends on the specific value $\text{rank}(E_8) + 1 = 9 = 3^2$ (composite, with $3 \mid h$).

10 The action

The complete Havelock field theory action on the spacetime $M_4 = \mathbb{R} \times (\mathbf{H}^2 \times_N S^1)$ is:

$$S = S_{\text{grav}} + S_{\text{CS}} + S_{\text{gauge}} + S_{\text{scalar}} + S_{\text{ferm}} + S_{\text{Yuk}}. \quad (10.1)$$

Gravitational sector. After KK reduction on S^1 , the 2+1D gravitational action is the Chern–Simons action:

$$S_{\text{CS}} = \frac{k}{4\pi} \int_{M_3} \text{Tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right), \quad k = \frac{c}{6} = 2b(N), \quad (10.2)$$

with A the $\text{SU}(2)$ (Euclidean) gauge field.

U(1) gauge sector.

$$S_{\text{gauge}} = -\frac{1}{4\alpha} \int d^4x \sqrt{-g} F_{\mu\nu} F^{\mu\nu}, \quad \alpha = \frac{6}{\pi c}. \quad (10.3)$$

Scalar sector (KK modes). N charged scalars Φ_m ($m = 0, \dots, N-1$) with KK masses $\mu_m = |m - N/2|$:

$$S_{\text{scalar}} = \int d^4x \sqrt{-g} \sum_{m=0}^{N-1} \left[|D_\mu \Phi_m|^2 - \mu_m^2 |\Phi_m|^2 \right], \quad (10.4)$$

where $D_\mu \Phi_m = (\partial_\mu - imA_\mu) \Phi_m$.

Fermion sector. N Dirac fermions Ψ_m with masses $\mu_m^f = |m - (N-1)/2|$:

$$S_{\text{ferm}} = \int d^4x \sqrt{-g} \sum_{m=0}^{N-1} \bar{\Psi}_m (i \not{D} - \mu_m^f) \Psi_m. \quad (10.5)$$

Yukawa sector.

$$S_{\text{Yuk}} = \int d^4x \sqrt{-g} \sum_{i,j} Y_{ij} \bar{\Psi}_i \Phi \Psi_j + \text{h.c.}, \quad (10.6)$$

with the selection rule $m_i - m_j + m_H \equiv 0 \pmod{N}$ (KK charge conservation on S^1).

All couplings are determined by $c = 12b(N)$.

The 4D effective Lagrangian. At energies $E \ll M_{\text{poly}}$, the KK modes (graviphoton, radion, and the KK tower) decouple, and the effective theory on $\mathbb{R}^{3,1}$ is:

$$\begin{aligned} \mathcal{L}_{4\text{D}} = & -\frac{1}{16\pi G_4} (R - 2\Lambda) && [\text{gravity, from boundary } T] \\ & -\frac{1}{4} \sum_a F_{\mu\nu}^a F^{a,\mu\nu} && [\text{SU}(3) \times \text{SU}(2) \times \text{U}(1), \text{ from boundary } J^a] \\ & + \bar{\Psi} (i \not{D} - YH) \Psi && [3 \text{ gen. fermions, } \mathbf{H}^2 \text{ radial profiles on } \mathbf{H}^2] \\ & + |D_\mu H|^2 - V(H) && [\text{Higgs, } v = 246 \text{ GeV, } \lambda_H = 0.122] \end{aligned} \quad (10.7)$$

All couplings in (10.7) are determined by $N = 7$ and M_P ; no free parameters remain.

Remark 10.1 (Unified 3+1D emergence from the boundary CFT). *The 3+1D theory is not assembled from separate mechanisms. It emerges from a single object: the boundary CFT at $c = 12b(N)$ on the torus $T^2 = \partial M$. The CS/WZW theorem encodes all bulk degrees of freedom in the boundary operators, organized by conformal weight:*

<i>Boundary operator</i>	<i>Weight</i>	<i>4D field</i>	<i>Origin</i>
<i>Stress tensor</i> $T(z)$	$(2, 0)$	<i>Graviton</i> $h_{\mu\nu}$	<i>Virasoro</i>
<i>WZW currents</i> $J^a(z)$	$(1, 0)$	<i>Gauge bosons</i> A_μ^a	<i>Kac–Moody</i>
<i>KK primaries</i> $\Phi_n(z)$	$(h_n, \bar{h}_n)$	<i>Matter fields</i>	<i>Fiber modes</i>
<i>Modular parameter</i> τ	—	<i>Cosmological</i> Λ	<i>Partition fn</i>

The graviton, gauge bosons, and matter are three sectors of this single boundary theory, not three independent constructions. The time direction emerges from the Wick rotation $t_E \rightarrow it_L$ of the Euclidean boundary CFT to Lorentzian signature, which is well-defined because the CFT is unitary (DHVW orbifold of a unitary WZW model). Lorentz invariance follows from the Einstein equations (Theorem 3.3).

11 UV completion

The theory is UV-complete in the following precise sense.

Non-perturbative definition. The boundary CFT is a $\mathbb{Z}_N$ orbifold theory at central charge $c = 12b(N)$, with known partition function, OPE coefficients, and conformal blocks. By the CS/WZW theorem [Witten, 1988], this boundary theory defines the bulk Chern–Simons theory at $k = c/6 = 2b(N)$ non-perturbatively. At leading order, $c = N(N+1) + O(\ln N)$; the subleading corrections $(-12 \ln 2 + 12(\ln N)/(N-1))$ encode the regularized vortex self-energy. The c -theorem is satisfied: the free-field UV limit has $c_{\text{UV}} = N(N+1)$, and $c_{\text{IR}} = 12b(N) < N(N+1)$ for all $N \geq 3$. There is a single central charge throughout: $c = 12b(N)$ enters both the Brown–Henneaux relation ($G = 3/(2c)$) and the WDW breathing-mode equation.

Super-renormalizability. After KK reduction to 1+1D, the superficial degree of divergence is $D = 2 - E$ (E = external legs). Only the 2-point function ($D = 0$) has a logarithmic divergence; all vertices with $E \geq 4$ are UV-finite. Two renormalization parameters suffice: the vortex core size a (UV cutoff for self-energy) and the central charge c (overall energy scale).

Physical observables. All physical observables. Havelock eigenvalues, partition function, couplings, cosmological constant, Weinberg angle, are exact (no perturbative approximation). The only divergences (self-energy at $d = 0$, KK zero-point sum) are renormalized by physical parameters within the framework.

The S^3 regime and E_8 (Paper V). The Seifert manifold $\mathbf{H}^2 \times_N S^1$ has Euler class $e = N/2$, giving the SM couplings at $K \leq 0$. For $K > 0$ (spatial S^3), the Hopf fibration $S^1 \rightarrow S^3 \rightarrow S^2$ has Euler class $e = 1$ (the first Chern class of $\mathcal{O}(1)$ on $\mathbb{CP}^1 = S^2$), fixing the CS level at $k = 1$. At this level, the full gauge group is E_8 with $1/g^2 = k + h^\vee = 31$ and WZW central charge $c = 248/31 = 8$ (exactly). The $E_8 \rightarrow$ SM breaking at $K = 0$ is a saddle-point transition driven by a Thurston geometry change ($S^3 \rightarrow \mathbb{R}^3$): the icosahedral ground state ($N = 12$, A_5 , E_8) ceases to exist as a finite-energy configuration, and the heptagonal ground state ($N = 7$, $\mathbb{Z}_7$, SM) takes over. No Higgs mechanism is involved. The 236 generators (E_8/SM , dimension $248 - 12$) have no finite-energy realization at $K = 0$. This is developed in full in Paper V.

KK spectrum and corrections. The KK tower on the Seifert manifold has mass-squared $M_{n,m}^2 = (\lambda_m + n^2) M_{\text{KK}}^2$ ($n \geq 0$, $1 \leq m \leq N-1$, $\lambda_m = f(m^*, N) - f(m, N)$). At $N = 11$ the lowest states are:

n	m	M/M_{KK}	Identity
0	5, 6	0	graviton (critical modes)
0	4, 7	1	lightest DM ($\lambda = 1$)
1	5, 6	1	first graviton KK
0	3, 8	$\sqrt{3}$	DM ($\lambda = 3$)
1	4, 7	$\sqrt{2}$	first DM ₁ KK
0	2, 9	$\sqrt{6}$	DM ($\lambda = 6$)
0	1, 10	$\sqrt{10}$	DM ($\lambda = 10$)

The lightest DM particle and the first graviton KK excitation are *mass-degenerate* at M_{KK} , a structural consequence of $\lambda_4 = 1 = 1^2$. The one-loop corrections from the full tower, summed via zeta regularization ($\sum_{n=1}^{\infty} n^{-2} = \pi^2/6$, $\sum_{n=1}^{\infty} \ln n = -\zeta'(0) = \frac{1}{2} \ln(2\pi)$), are:

Observable	$M_{\text{KK}} = 50 \text{ TeV}$	$M_{\text{KK}} = 300 \text{ TeV}$
$\delta G/G$	2.5×10^{-7}	7×10^{-9}
$\delta g^2/g^2$	2.5×10^{-7}	7×10^{-9}
$\delta m_H^2/m_H^2$	1.5×10^{-7}	4×10^{-9}

All corrections are 10^{-7} – 10^{-9} , six to eight orders of magnitude below the geometric-tier accuracy and four to six orders below the algebraic tier. The boundary CFT controls the UV; the KK corrections are finite, calculable effects in the IR effective theory.

12 One-loop corrections

The Chern–Simons non-renormalization theorem guarantees that the CS level receives *at most* a one-loop correction; higher loops are forbidden by large-gauge-transformation invariance.

Proposition 12.1 (One-loop structure).

1. **Coupling lock (all-orders exact).** The ratio $\alpha/(8\pi G) = 1/(2\pi^2)$ is algebraically c -independent: both G and α are proportional to $1/c$, so their ratio cancels c . The one-loop shift $c \rightarrow c - 3N$ (N Dirac fermions, each contributing $c = -3$ from the gravitational anomaly: -1 from the Dirac field -2 from the parity anomaly in odd dimensions) does not affect the ratio. This is exact at all orders.
2. **CS level shift.** Each massive 3D Dirac fermion shifts the CS level by $\text{sgn}(\mu_m)/2$ (parity anomaly). The KK spectrum $\mu_m = (m + \frac{1}{2} - N/2)/R$ is antisymmetric, so the massive modes cancel pairwise; for odd N , the single zero mode at $m = (N-1)/2$ contributes $\pm 1/2$ (sign from fiber orientation). The physical level is $k_{\text{phys}} = c/6 - 1/2$ (odd N) or $c/6$ (even N). All downstream quantities depend only on the fractional part $\kappa = \text{frac}(k_{\text{phys}}) = 0.096$ (at $N = 7$), which is unaffected by the integer shift $-(N-1)/2$. No further renormalization occurs beyond one loop.

3. **Cosmological constant.** The 3D Coleman–Weinberg correction $\Delta\Lambda = -(1/12\pi)(\sum_b \mu_m^3 - \sum_f |\mu_m^f|^3)$ is a finite sum. At $N = 11$: $\Delta\Lambda/\Lambda_{\text{tree}} = -9.1\%$.
4. **Fermion mass ratios (quantum stable).** The one-loop self-energy from CS exchange gives a universal mass shift $\delta m/m = -1/(4\pi k)$ ($< 1\%$ for $N \geq 7$), identical for all KK modes. Mass ratios and Yukawa texture zeros are unchanged.
5. **Weinberg angle and the polygon scale.** The tree-level prediction $\sin^2 \theta_W = 3/11$ matches the experimental 0.2312 at M_Z via standard SM running for

$$M_{\text{poly}} \approx 300 \text{ TeV} \quad (10^{5.5} \text{ GeV}). \quad (12.1)$$

At $M_{\text{poly}} = 50 \text{ TeV}$ the prediction is $\sin^2 \theta_W(M_Z) = 0.240$ (3.9% above experiment); $O(1)$ threshold corrections at the polygon scale accommodate the 50–500 TeV range.

13 The fermion mass structure

This section derives the fermion spectrum from the Seifert geometry. Every result below follows from the $\mathbb{Z}_7$ orbifold structure, the $\mathbf{H}^2$ radial profiles, and the BF instanton on $\mathbf{H}^2$; no structural identifications are needed beyond those already established in §8.6. The results fall into three tiers: *Exact* (no adjustable parameters): the Yukawa texture zeros (§13.1–§13.3), the CKM phase $\delta = 69.3^\circ$ (§13.8), and sector decoupling (§13.6). *Warp factor derived*: the mass hierarchy (§13.5) and tree-level CKM (§13.7) depend on the warp factor σ , which is determined by N alone (§13.4): $\sigma_{\text{CKM}} = N-2 = 5$, $\sigma_{\text{mass}} = (N-2)\sqrt{N} = 5\sqrt{7}$. All mass ratios and mixing angles are then fixed. *Non-perturbative*: the orbifold image suppression (§??) gives $|V_{cb}| = 0.043$.

The derivation chain follows the gauge-sector structure of §8.6, extended to fermions. The only additional input is σ , which is derived from N (§13.4).

13.1 The Yukawa texture from KK charge conservation [exact]

The Yukawa coupling Y_{ij} between fermion pairs i, j and the Higgs (pair 3, modes 3 and 4) is nonzero iff $m_i - m_j + m_H \equiv 0 \pmod{7}$. Evaluating for all pairs:

$$Y = \begin{pmatrix} 0 & Y_{12} & Y_{13} \\ Y_{21} & Y_{22} & 0 \\ Y_{31} & 0 & 0 \end{pmatrix}, \quad (13.1)$$

with four texture zeros at $(1, 1)$, $(2, 3)$, $(3, 2)$, $(3, 3)$, determined entirely by the arithmetic of $\mathbb{Z}/7\mathbb{Z}$.

13.2 The 2+1 mass pattern and the exact tree-level ratio [exact]

The matrix $YY^\dagger$ is block-diagonal: a 2×2 block (pairs 1,2) and a 1×1 block (pair 3). The eigenvalues of the 2×2 block (in the large- σ limit where the $\mathbf{H}^2$ radial profiles saturate) are $\lambda_\pm = (5 \pm \sqrt{13})/2$, giving the exact tree-level mass ratio

$$\left. \frac{m_1}{m_2} \right|_{\text{tree}} = \sqrt{\frac{5 + \sqrt{13}}{5 - \sqrt{13}}} \approx 2.48, \quad (13.2)$$

while the lightest mass $m_3 \propto e^{-\sigma/2}$ is exponentially suppressed by the warp factor σ .

13.3 Up–down splitting from the $N = 4$ isospin [exact]

The $SU(2)$ doublet structure from the $N = 4$ sector gives different KK masses for up-type ($\mu_4 = 1/2$) and down-type ($\mu_4 = 3/2$) fermions. The combined bulk mass $c_{k,I} = \sqrt{\mu_{7,k}^2 + \mu_{4,I}^2}$ separates the two types, with down-type having larger c (more UV-localized, hence lighter).

13.4 The warp factor tower [derived from N]

The RS warp factor σ , previously treated as a free parameter, is determined by N alone. Three σ values appear in the fermion sector; all derive from the polygon order:

Parameter	Value	Formula	Origin
σ_{geo}	7	N	KK warp factor (polygon order)
σ_{CKM}	5	$N-2$	nonzero Yukawa entries
σ_{mass}	$5\sqrt{7}$	$(N-2)\sqrt{N}$	CKM warp $\times \sqrt{N}$

(i) $\sigma_{\text{geo}} = N$ (derived from the Euler class). The Seifert connection on $\mathbf{H}^2 \times_N S^1$ has first Chern number $c_1 = e = N/2$ (Proposition 3.1). The $\mathbb{Z}_N$ orbifold divides the base into N sectors, each carrying flux $c_1/N = 1/2$ (a Chern half-unit). The RS warp factor for a KK mode with unit charge traversing one sector is

$$\sigma_{\text{image}} = e \times \frac{\text{Area}_{\text{image}}}{\text{Area}_{\text{total}}} \times 2 = \frac{N}{2} \times \frac{1}{N} \times 2 = 1,$$

where the factor 2 arises because the RS extra dimension is $S^1/\mathbb{Z}_2$, not S^1 : the physical range is $[0, \pi R_c]$ (the fundamental domain of the $\mathbb{Z}_2$ orbifold), which is *half* the full circle $[0, 2\pi R_c]$. The total holonomy $\oint A = 2\pi c_1$ traverses the full circle, but the physical warp traverses the fundamental domain, giving $\sigma = (2\pi c_1)/\pi = 2c_1 = 2e$ (?). With N sectors each contributing one e -fold: $\sigma_{\text{geo}} = N$. The polygon scale is therefore $M_{\text{poly}} = v e^N = v e^7 = 270 \text{ TeV}$ (consistent with the polygon scale $\sim 300 \text{ TeV}$ from the Weinberg angle running; the 1.5% residual is a threshold correction).

(ii) $\sigma_{\text{CKM}} = N - 2$. The $\mathbb{Z}_7$ charge conservation $m_i - m_j + m_H \equiv 0 \pmod{7}$ produces a 3×3 Yukawa texture with exactly 4 zeros and $5 = N - 2$ nonzero entries (13.1). Each nonzero Yukawa is one RS-suppressed coupling, contributing one e -fold to the CKM hierarchy. The count $N - 2$ arises because the $N - 1 = 6$ KK modes include one critical pair (the Higgs at $\lambda = 0$) that does not contribute to the warp; the remaining $N - 1 - 1 = N - 2 = 5$ modes each contribute one unit.

(iii) $\sigma_{\text{mass}} = (N - 2)\sqrt{N}$ (derived from the Dirac eigenvalue on the Seifert manifold). The CKM mixing angles probe the S^1 fiber alone (KK mass $\Lambda_{S^1} = m/R$, one quantum number). The fermion mass hierarchy probes the *full* Seifert geometry: the Dirac operator on $\mathbf{H}^2 \times_N S^1$ has eigenvalue $\Lambda_{\text{Seifert}}^2 = f(m^*, N) + m^2$, where $f(m^*, N)$ is the Havelock Casimir of the critical mode on $\mathbf{H}^2$ and m is the KK charge on S^1 . The ratio of effective warp factors is

$$\frac{\sigma_{\text{mass}}}{\sigma_{\text{CKM}}} = \frac{\Lambda_{\text{Seifert}}}{\Lambda_{S^1}} = \sqrt{\frac{f(m^*, N) + m^2}{m^2}} = \sqrt{f(m^*, N) + 1} \quad \text{at } m = 1.$$

At $N = 7$: $f(3, 7) = 6$, giving $\sqrt{6+1} = \sqrt{7} = \sqrt{N}$.

This identity is *specific to* $N = 7$. At the critical mode $m^* = (N-1)/2$:

$$f\left(\frac{N-1}{2}, N\right) + 1 = \frac{N^2-1}{8} + 1 = \frac{N^2+7}{8}.$$

This equals N if and only if $N^2 - 8N + 7 = 0$, i.e. $(N-1)(N-7) = 0$. The only non-trivial solution is $N = 7$: the same value selected by Havelock stability and von Staudt–Clausen. Therefore $\sigma_{\text{mass}} = \sigma_{\text{CKM}} \times \sqrt{N} = (N-2)\sqrt{N} = 5\sqrt{7} \approx 13.23$.

13.5 The fermion mass hierarchy [derived: $\sigma = (N-2)\sqrt{N}$]

The tree-level texture (13.1) has $Y_{33} = 0$ (the charge conservation $3+3+3 = 9 \not\equiv 0 \pmod{7}$ forbids a direct b -quark Yukawa). At tree level, m_b comes from $Y_{31} = f_3 f_1$, which involves the most UV-localized profile f_1 . The Dirac zero-mode profile on the Seifert manifold is $\psi_c(\rho) = N_c (\sinh \rho)^{c-1/2}$ (from the KK-reduced Dirac equation with warp factor $A(\rho) = \ln(\sinh \rho)$). In the holographic regime $\rho \gg 1$, $\sinh \rho \approx e^\rho/2$ and the profile reduces to the Randall–Sundrum form $f_k = \sqrt{(2c_k-1)/(e^{(2c_k-1)\sigma}-1)}$ at $\sigma = \sigma_{\text{mass}} = (N-2)\sqrt{N} = 5\sqrt{7}$, the derived warp factor from the Seifert geometry (§13.4). The profile suppresses f_1 exponentially ($c_1 = 2.50$), giving the tree-level ratio $m_s/m_b \approx 0.63$.

The BF instanton resolves this. The same BF-crossing instanton that generates $\delta_{\text{CKM}} = 69.3^\circ$ (Theorem 13.1) carries $\Delta q = 2$ units of $\mathbb{Z}_7$ charge and generates an effective Y_{33} :

$$Y_{33}^{\text{inst}} = \mathcal{K}^2 f_3^2, \quad \mathcal{K} = 0.548 \text{ (eq. (6.2))}. \quad (13.3)$$

Since $\mathcal{K}^2 f_3 \gg f_1$ at the physical warp factor ($\mathcal{K}^2 f_3/f_1 \approx 260$ at $\sigma = 7.1$), the instanton contribution dominates: $m_b \propto Y_{33}^{\text{inst}} = \mathcal{K}^2 f_3^2$. The s -quark mass comes from the tree-level $Y_{22} = f_2^2$ (charge-conserving, no instanton needed). The mass ratio is therefore

$$\frac{m_s}{m_b} = \frac{f_2^2}{\mathcal{K}^2 f_3^2} = \frac{1}{\mathcal{K}^2} \left(\frac{v}{M_{\text{poly}}} \right)^{2\Delta c}, \quad (13.4)$$

where $\Delta c = c_2 - c_3$ is the conformal-dimension gap between generations 2 and 3, with $c_k = \sqrt{\mu_{7,k}^2 + \mu_4^2}$ and $\mu_{7,k} = |k - (N-1)/2|$ (the pair-centered KK mass). At tree level $\Delta c_0 = \sqrt{1+9/4} - 3/2 = 0.303$. The Seifert fiber connection shifts the effective KK charge by $e/c = (N/2)/(12b(N))$ per unit of angular momentum (the same Euler class $e = N/2$ that drives parity violation in §8.1). Combined with the $O(1/c)$ BO anharmonic correction, the conformal-dimension gap becomes

$$\Delta c = \Delta c_0 + \frac{\mu_{7,2}}{c_2} \cdot \frac{1+e}{c} = 0.303 + 0.048 = 0.351. \quad (13.5)$$

The two corrections (Euler class and BO anharmonic) are additive because they act on different sectors: the Euler class shifts the KK charge (a geometric, fiber-connection effect), while the BO anharmonic shifts the radial potential (a spectral effect from the W_N quartic coupling). Their cross-term is $O(\mu_{7,2} e/c^3) = O(10^{-5})$, negligible at the precision of (13.5). Standard SM two-loop running (with $b_1 = 41/10$, $b_2 = -19/6$, and the two-loop matrix b_{ij} of Machacek and Vaughn 1984) gives $\sin^2 \theta_W = 3/11$ at $M_{\text{poly}} = 294 \text{ TeV}$.⁴

⁴The two-loop SM β -functions [Machacek and Vaughn, 1984] shift M_{poly} from 274 (one-loop) to 294 TeV (two-loop); see Code Availability.

The resulting mass ratio: $m_s/m_b = 0.023$ (observed 0.024, 4% match). One calibration input (Δm_{atm}^2) pins M_{poly} within the predicted range; all other quantities are computed: $\mathcal{K}$ is fixed by the CKM phase, Δc by the KK spectrum, and M_{poly} by the Weinberg angle (with two-loop corrections).

The tree-level CKM has $|V_{cb}|_{\text{pert}} = 0.092$ ($2.2\times$ the observed 0.042). The non-perturbative CKM (§??) resolves this separately: J -conservation combined with the scattering phase $\delta = 69.3^\circ$ redistributes the mixing angles, giving $|V_{cb}|_{\text{np}} = 0.044$ (6% above observed).

The d -quark mass from CKM mixing (Gatto relation). The bare Yukawa eigenvalue $m_d^0 \propto f_1^2$ is doubly UV-suppressed and negligible ($m_d^0 \approx 0.002 \text{ MeV}$ at $\sigma = 7.1$). The physical d -quark mass comes from CKM mixing with the s quark:

$$m_d \approx |V_{us}|^2 m_s, \quad \frac{m_d}{m_s} = \sin^2 \theta_C. \quad (13.6)$$

With $|V_{us}| = 0.230$ (the tree-level Cabibbo angle): $m_d/m_s = 0.056$ (observed: 0.049, 14% off). This is the Gatto relation [Gatto et al., 1968]: no new parameter enters. The three down-type mass ratios are then: $m_s/m_b = 0.024$ (eq. (13.4)), $m_d/m_s = 0.056$ (eq. (13.6)), $m_d/m_b = 0.0013$ (product; observed 0.0011).

13.5.1 The u -quark mass and the broad Higgs

The up-type conformal dimensions are $c_t = \sqrt{0+(1/2)^2} = 1/2$ (exactly BF-marginal: the top quark is conformally coupled), $c_c = \sqrt{1+1/4} = 1.12$, $c_u = \sqrt{4+1/4} = 2.06$. The BO ground state (Higgs profile) has width $\sigma_H = 1/\sqrt{V''(\rho^*)} = 2.74$, which exceeds $\rho^* = 1.73$: the Higgs wavefunction extends to $\rho \approx 0$, sampling the unsuppressed region where all RS profiles are comparable. The effective (1,2) mixing angles from the overlap integral $\theta_{12}^I = \int f_1^I |\Psi_H|^2 \sinh \rho d\rho / \int f_2^I |\Psi_H|^2 \sinh \rho d\rho$ are $\theta_{12}^{\text{down}} = 0.28$ and $\theta_{12}^{\text{up}} = 0.032$ (the up-sector mixing is $9\times$ smaller because $\Delta c_{\text{up}} = 0.94 > \Delta c_{\text{down}} = 0.70$). This gives simultaneously: $|V_{us}| \approx |\theta_d - \theta_u| = 0.25$ (observed 0.224), $m_d \approx \theta_d^2 m_s = 7.6 \text{ MeV}$ (observed 4.7), $m_u \approx \theta_u^2 m_c = 1.3 \text{ MeV}$ (observed 2.2). The 40–60% discrepancies on m_u , m_d reflect the sensitivity of light-quark masses to the Higgs profile tail and QCD running to 2 GeV; the qualitative prediction $m_u < m_d \ll m_s$ is robust and parameter-free.

13.6 Sector decoupling [exact]

The $N = 7$ and $N = 8$ polygon sectors are decoupled by the mod-56 arithmetic of the common overfield $\mathbb{Q}(\zeta_{56})$: the cross-sector Yukawa coupling requires $8(m_1 + m_2) \equiv 28 \pmod{56}$, which has no integer solution ($\gcd(8, 56) = 8$ does not divide 28). The texture zeros of the $N = 7$ Yukawa are topologically protected.

13.7 CKM mixing matrix [derived: $\sigma = N-2$]

The CKM matrix $V = U_L^{(\text{up})\dagger} U_L^{(\text{dn})}$ is extracted from the orbit-based Yukawa matrices using the *left* rotation: U_L diagonalizes $YY^\dagger$ (not $Y^\dagger Y$). The Bernoulli instanton phases $\alpha_L = 1/7$, $\alpha_R = 5/42$, $\alpha_H = 2/21$ enter each Yukawa entry through $\exp(2\pi i(\alpha_L m_L + \alpha_R m_R + \alpha_H m_H)/N)$. Predicted at $\sigma = 5$:

$$|V| \approx \begin{pmatrix} 0.973 & 0.230 & 0.088 \\ 0.230 & 0.971 & 0.043 \\ 0.088 & 0.043 & 0.995 \end{pmatrix} \quad \text{Observed:} \quad \begin{pmatrix} 0.974 & 0.224 & 0.004 \\ 0.224 & 0.974 & 0.042 \\ 0.008 & 0.039 & 0.999 \end{pmatrix}$$

The Cabibbo angle $\theta_C = 13.7^\circ$ (observed 13.0° , 5% off). The hierarchy $|V_{us}| \gg |V_{cb}| \gg |V_{ub}|$ is predicted.

13.8 The CKM phase [exact]

Theorem 13.1 (CKM phase from the $\mathbb{Z}_7$ Gauss sum). *The CKM CP-violating phase is*

$$\delta_{\text{CKM}} = \arctan \sqrt{7} = 69.30^\circ, \quad (13.7)$$

with $\sin^2 \delta = N/(N+1) = 7/8$ exactly (Corollary 8.5). The CKMfitter 2024 global average is $\gamma = 65.9^\circ \pm 3.5^\circ$; the prediction is 1.0σ from the central value.

Proof. The derivation proceeds in five steps from $N = 7$.

Step 1: Von Staudt–Clausen selects $N = 7$. By Theorem 8.6, $N \mid \text{denom}(B_{N-1})$ iff N is prime. At $N = 7$: $\text{denom}(B_6) = 42 = 2 \cdot 3 \cdot 7$, so $7 \mid 42$. Combined with Havelock stability ($\lambda_{\min}(7) = 0$, $\lambda_{\min}(8) < 0$), this selects $N = 7$.

Step 2: Quadratic residues = Frobenius orbits. The quadratic residues mod 7 are $\text{QR} = \{1, 2, 4\}$ (since $1^2 \equiv 1$, $2^2 \equiv 4$, $3^2 \equiv 2 \pmod{7}$). The non-residues are $\text{QNR} = \{3, 5, 6\}$. These partition $(\mathbb{Z}/7\mathbb{Z})^*$ into two orbits of size 3, identifying three up-type and three down-type generations.

Step 3: The partial Gauss sum. $G(\text{QR}) = \omega + \omega^2 + \omega^4 = (-1 + i\sqrt{7})/2$, where $\omega = e^{2\pi i/7}$. (This is the *partial* sum over quadratic residues $\text{QR} = \{1, 2, 4\}$ only, not the full quadratic Gauss sum $\sum_{k=0}^6 e^{2\pi i k^2/7} = i\sqrt{7}$, which has argument $\pi/2$.) For primes $p \equiv 3 \pmod{4}$: $G(\text{QR}) + G(\text{QNR}) = -1$ and $G(\text{QR}) - G(\text{QNR}) = i\sqrt{7}$, giving $G(\text{QR}) = (-1 + i\sqrt{7})/2$ with $\arg G(\text{QR}) = \pi - \arctan \sqrt{7} = 110.7^\circ$.

Step 4: The sector amplitude. The Yukawa three-point function in the $\mathbb{Z}_7$ orbifold sums over all orbifold sectors with *equal* weight: $Y = (1/|G|) \sum_{m=0}^{N-1} Y_m$ (this is the definition of the orbifold path integral; [Dixon et al. 1985](#)). Three facts fix the relative coefficients:

1. *Orbifold averaging.* Each sector m contributes Y_m with coefficient $1/|G|$. No sector is weighted more than any other (the $\mathbb{Z}_N$ orbifold projection).
2. *Identity operator.* The untwisted sector ($m = 0$) inserts the identity. In any CFT, the identity has unit OPE coefficient: $\langle O_1(z) O_2(0) \rangle = C_{12}^1 = 1$ for canonically normalized operators (?, eq. 2.6.14). Therefore $Y_0 = 1 \times Y_{\text{tree}}$.
3. *Twist-field normalization.* Each twisted sector m inserts the twist field σ_m . With the DHVW normalization $\langle \sigma_m \sigma_{-m} \rangle = 1$, the three-point function gives $Y_m = \omega^m \times Y_{\text{tree}}$ (the twist eigenphase ω^m is the $\mathbb{Z}_N$ charge of the operator; the OPE coefficient is 1 by the unit two-point function).

Combining: the up-type amplitude (summing over QR sectors $m \in \{1, 2, 4\}$) is

$$\alpha_2 = \underbrace{1}_{\text{identity (Step 2)}} + \underbrace{G(\text{QR})}_{\text{twist (Step 3)}} = \frac{1 + i\sqrt{7}}{2},$$

where the overall factor $Y_{\text{tree}}/|G|$ is real and cancels in the phase. The magnitude $|Y_m|$ is independent of m because the $\mathbb{Z}_N$ symmetry permutes the twisted sectors: the orbifold action maps sector m to sector $m+1$, so all sectors have identical OPE coefficient magnitudes. Only the phase ω^m distinguishes them. No coefficient is a free parameter: each follows from the orbifold construction and CFT operator normalization. The down-type amplitude is $\bar{\alpha}_2 = (1 - i\sqrt{7})/2$ (since $G(\text{QNR}) = \overline{G(\text{QR})}$ for $p \equiv 3 \pmod{4}$).

Step 5: The CKM phase. $\delta_{\text{CKM}} = \arg(\alpha_2) = \arctan(\sqrt{7})$. CP violation requires $N \equiv 3 \pmod{4}$ (so the Gauss sum is complex); $7 \equiv 3 \pmod{4}$. The class number $h(-7) = 1$ (Heegner) ensures the CM point is unique. $\square$

The Jarlskog invariant with PDG mixing angles: $J = s_{12} s_{23} s_{13} c_{12} c_{23} c_{13}^2 \sin \delta = 3.14 \times 10^{-5}$ (observed: 3.08×10^{-5} , 2% match). The CP violation is near-maximal: $\sin \delta = \sqrt{7/8} = 0.935$.

13.9 The orbit-based CKM matrix

The CKM matrix is extracted from the orbit-based Yukawa using the LEFT rotation: $V = U_L^{(\text{up})\dagger} U_L^{(\text{dn})}$, where U_L diagonalizes $YY^\dagger$ (not $Y^\dagger Y$). The instanton phases are $\alpha_L = 1/7$, $\alpha_R = 5/42$, $\alpha_H = 2/21$ from the pattern $(N-k)/\text{denom}(B_6)$ for $k = 1, 2, 3$. The ratio $\alpha_L/\alpha_R = 6/5 = M_2$ (Corollary 8.4).

Theorem 13.2 (Instanton correction for V_{ub}). $|V_{ub}| = |V_{ub}|_{\text{tree}} \times \mathcal{K}^{N-1}(1 + \mathcal{K} + O(\mathcal{K}^2))$, where $\mathcal{K} = e^{-2\pi k_{\text{frac}}} = 0.548$ is the instanton fugacity. At leading order (LO): $|V_{ub}| = 0.0024$ (35% below PDG). At next-to-leading order (NLO), the single-instanton correction $(1 + \mathcal{K})$ gives $|V_{ub}| = 0.0037$ (2% match to PDG). The correction applies only to V_{ub} , not to V_{us} or V_{cb} .

Proof. The Yukawa texture (13.1) gives $(YY^\dagger)_{02} = 0$ (since $Y_{00} = Y_{21} = Y_{22} = 0$), so V_{ub} arises only from the indirect path $1 \rightarrow 2 \rightarrow 3$ through generation 2. The instanton winding for this path: $w_{\text{up}} = |1 - 4| = 3 = (N-1)/2$ (max gap in QR), $w_{\text{dn}} = |6 - 3| = 3$ (max gap in QNR), $w_{\text{total}} = 6 = N-1$. The leading instanton amplitude is $\mathcal{K}^{N-1}$ (one tunneling event per winding unit). The NLO correction $(1 + \mathcal{K})$ accounts for the sub-leading instanton with one additional winding unit ($w = N$ instead of $w = N-1$); this is the standard dilute-gas correction to the semiclassical tunneling amplitude. Direct-mixing elements $(YY^\dagger)_{01} \neq 0$ and $(YY^\dagger)_{12} \neq 0$, so V_{us} and V_{cb} are tree-level and uncorrected. $\square$

The five CKM predictions (at $\sigma_{\text{CKM}} = 5$):

Observable	Prediction	PDG	Match
δ	$\arctan \sqrt{7} = 69.3^\circ$	$65.9 \pm 3.5^\circ$	1.0σ
$ V_{us} $	0.230	0.225	2%
$ V_{cb} $	0.043	0.042	1%
$ V_{ub} $ (LO: $\mathcal{K}^6$)	0.0024	0.00365	35%
$ V_{ub} $ (NLO: $\mathcal{K}^6(1+\mathcal{K})$)	0.0037	0.00365	2%
J (NLO)	3.0×10^{-5}	3.08×10^{-5}	3%

13.10 PMNS neutrino mixing

The PMNS matrix arises from the *pair structure* of $(\mathbb{Z}/7\mathbb{Z})^*$. The CKM uses the quadratic residue orbit $\text{QR} = \{1, 2, 4\}$; the PMNS uses the three pair modes $\{1, 6\}$, $\{2, 5\}$, $\{3, 4\}$.

Theorem 13.3 (Pair symmetry forces maximal atmospheric mixing). *In a neutrino mass matrix of the form $m_\nu = C^T \text{diag}(w) C$, where $C_{km} = \cos(2\pi km/N)$ is the pair cosine matrix ($k, m \in \{1, \dots, (N-1)/2\}$) and w is any real weight vector, the atmospheric mixing angle $\theta_{23} = \pi/4$ is independent of w .*

Proof. By character orthogonality, $\sum_{k=1}^{(N-1)/2} \cos^2(2\pi k \cdot 2/N) = \sum_{k=1}^{(N-1)/2} \cos^2(2\pi k \cdot 3/N) = (N-2)/4$, since both reduce to $\frac{1}{2} \left(\frac{N-1}{2} + \frac{1}{2} \sum_k \cos(4\pi km/N) \right) = \frac{1}{2} \left(\frac{N-1}{2} - \frac{1}{2} \right) = \frac{N-2}{4}$. Columns 2 and 3 of C have identical norms. Since $m_\nu = C^T \text{diag}(w) C$, the 2–3 block of m_ν inherits this symmetry: $(m_\nu)_{22}$ and $(m_\nu)_{33}$ receive equal contributions from every weight w_k (the difference $\sum_k w_k [\cos^2(2\pi k \cdot 2/N) - \cos^2(2\pi k \cdot 3/N)] = 0$). Since C and w are both real, $m_\nu = C^T \text{diag}(w) C$ is real symmetric. In the 2–3 block, the diagonal entries are equal (proved above) and the off-diagonal entry $(m_\nu)_{23} = \sum_k w_k \cos(2\pi k \cdot 2/N) \cos(2\pi k \cdot 3/N)$ is real. For a real symmetric 2×2 matrix with equal diagonal entries, the eigenvectors are $(1, \pm 1)/\sqrt{2}$, giving $\theta_{23} = \pi/4$ regardless of the off-diagonal value and the weights w . $\square$

The pair symmetry $m \leftrightarrow N-m$ forces $\theta_{23} = 45^\circ$ at leading order. The deviation from maximality arises when this symmetry is broken by the modular form hierarchy. Evaluating the three cusp forms of $\Gamma(7)$ (the Klein quartic holomorphic differentials) at the CM point $\tau_0 = (1+i\sqrt{7})/2$ gives $|x| \gg |y| \gg |z|$, which shifts θ_{23} from 45° toward $\sim 49^\circ$.

The Havelock Casimir appears as the leading q -power of the Klein quartic coordinates: $f_a(\tau) = q^{f(a,N)/N} \prod_{n \geq 1} (1 - q^{Nn - (N-a)})(1 - q^{Nn - a})$, where $f(a, N) = a(N-a)/2$ is the Havelock Casimir. This connects the modular form structure directly to the stability eigenvalue backbone.

Conjecture 13.4 ($\mathbb{Z}_7$ PMNS fractions). *The three PMNS mixing angles are simple algebraic functions of $N = 7$ alone:*

$$\sin^2(2\theta_{12}) = \frac{N-1}{N} = \frac{6}{7}, \quad (13.8)$$

$$\sin^2 \theta_{23} = \frac{N+1}{2N} = \frac{4}{7}, \quad (13.9)$$

$$\sin^2 \theta_{13} = \frac{1}{N^2-1} = \frac{1}{48}. \quad (13.10)$$

Equivalently, $\cos(2\theta_{23}) = -1/N$ and $1 - \sin^2(2\theta_{12}) = 1/N$.

The predictions and comparison with PDG:

Angle	Prediction	PDG	Pull
θ_{12}	33.9°	$33.41 \pm 0.79^\circ$	0.6σ
θ_{23}	49.1°	$49.0 \pm 1.3^\circ$	0.1σ
θ_{13}	8.3°	$8.54 \pm 0.15^\circ$	1.6σ

All three angles are within 2σ of PDG central values. For comparison, tribimaximal mixing (TBM) gives $\theta_{12} = 35.3^\circ$ (2.4σ), $\theta_{23} = 45^\circ$ (3.1σ), $\theta_{13} = 0^\circ$ (excluded at $> 10\sigma$).

Remark 13.1 (Derivation status). *Theorem 13.3 establishes $\theta_{23} = 45^\circ$ from character orthogonality. The Klein quartic modular forms at the CM point $\tau_0 = (1+i\sqrt{7})/2$ break the pair symmetry and shift θ_{23} toward 49° (numerically verified: 47.8° with F_3 charged leptons and the $\text{PSL}(2, 7)$ Clebsch–Gordan Majorana texture). The exact $\mathbb{Z}_7$ fraction formulas in Conjecture 13.4 match the numerical trend but their complete group-theoretic derivation (in particular the θ_{12} prediction) remains open. The full derivation likely requires exact modular form arithmetic at the CM point using the $\text{PSL}(2, 7)$ representation theory of $S_2(\Gamma(7))$.*

The CP phase prediction $\delta_{\text{CP}}^{\text{PMNS}} = \arctan \sqrt{7} = 69.3^\circ$ (from the same Gauss sum as the CKM) is in tension with the current T2K/NOvA preference for $\delta_{\text{CP}} \approx -90^\circ$; DUNE and Hyper-Kamiokande will provide a decisive test.

14 Consistency checks

14.1 Gravitational consistency

Electroweak scale from the Planck mass. The one dimensionful input is M_P (equivalently G_N). The hierarchy formula (Paper V, Theorem 4.1) *predicts* the electroweak scale:

$$v = M_P / \exp(\mathcal{H}_7), \quad \mathcal{H}_7 = 2S_{\text{BO}}(7) + \Delta\varepsilon \ln \varepsilon_7 + \frac{1}{2} \ln(c_{11}/(24\pi^2)) = 38.46, \quad (14.1)$$

where $S_{\text{BO}}(7) = 18.27$ is the WKB tunneling action of the $N = 7$ breathing mode (from $b(7)$ and $f(3,7)$), $\varepsilon_7 = 8 + 3\sqrt{7}$ is the fundamental unit of $\mathbb{Z}[\sqrt{7}]$ (from the Pell equation), $\Delta\varepsilon = 0.80$ is the WDW eigenvalue gap, and $c_{11} = 12b(11)$. Every component is computed from $b(N)$ and the Pell structure; no Standard Model input is used. This gives $v = 245.9 \text{ GeV}$ (observed: 246.2 GeV , 0.1% match). The hierarchy $M_P/v \approx 10^{17}$ is explained, not assumed.

Proposition 14.1 (4D graviton propagator). *The long-range 4D graviton propagator is the standard linearized GR result:*

$$\langle h_{ij}^{(4)}(k) h_{kl}^{(4)}(-k) \rangle = \frac{16\pi G_4}{k_4^2} \mathcal{P}_{ijkl}^{\text{TT}}, \quad (14.2)$$

with 2 tensor polarizations $(+, \times)$ and no scalar or vector admixture at $k \ll M_{\text{poly}}$.

Proof. Four steps, each using a standard result.

Step 1 (Boundary $\langle TT \rangle$). In the orbifold CFT at central charge c , the stress tensor 2-point function is fixed by conformal invariance: $\langle T_{ij}(k) T_{kl}(-k) \rangle = (c/24\pi) |k|^3 \mathcal{P}_{ijkl}^{\text{TT}}$, where $\mathcal{P}^{\text{TT}}$ is the transverse-traceless projector. This follows from conservation ($k^i \langle T_{ij} \rangle = 0$), tracelessness ($\delta^{ij} \langle T_{ij} \rangle = 0$), and conformal dimension 2.

Step 2 (Brown–Henneaux map). The normalizable bulk metric perturbation in AdS_3 satisfies $h_{ij}^{(2)}(k) = (16\pi G_3/c) \langle T_{ij}(k) \rangle$ [Brown and Henneaux, 1986]. This is a *tensor* equation: h inherits the $\mathcal{P}^{\text{TT}}$ structure of T .

Step 3 (KK mode sum). The boundary $\mathbb{R} \times S_\theta^1$ lifts on S_φ^1 (circumference $2\pi R = 2\pi/M_{\text{poly}}$) to $\mathbb{R} \times T^2$. The KK mode sum $G(k_3, x_\varphi) = \sum_n G_{2\text{D}}(k_3, m_n = n/R) e^{inx_\varphi/R}$ converts the 2D propagator to a 4D one: in the regime $k \gg 1/R$, the sum yields $G^{(4\text{D})} = G_3/(k_4^2 \times 2\pi R) = G_4/k_4^2$. The $\mathcal{P}^{\text{TT}}$ projector is *preserved*: S_φ^1 is orthogonal to the TT projection directions (which act on the $\mathbf{H}^2$ spatial coordinates).

Step 4 (Assembly). Combining Steps 1–3: $\langle h_{ij}^{(4)} h_{kl}^{(4)} \rangle = (16\pi G_4/k_4^2) \mathcal{P}_{ijkl}^{\text{TT}}$, with $G_4 = G_3/(2\pi R) = 1/M_P^2$ (matching the hierarchy formula). In position space, the standard Fourier transform $\int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{r}}/k^2 = 1/(4\pi r)$ yields Newton’s $1/r$ potential (Corollary 14.2). $\square$

The massive KK modes (graviphoton, radion) contribute additional DOF at $k > M_{\text{poly}}$ but are Yukawa-suppressed ($e^{-r/R} \approx 0$ at $R \sim 10^{-21} \text{ m}$) at macroscopic distances. Gravitational radiation at LIGO has tensor polarization $(+, \times)$, matching observations.

Corollary 14.2 (Newton’s law from the Seifert geometry). *The static gravitational potential between two masses m_1, m_2 at separation r on the Seifert spacetime is*

$$V(r) = -\frac{G_N m_1 m_2}{r} \quad (R \ll r \ll \ell), \quad (14.3)$$

with $G_N = G_4 = G_3/(2\pi R)$ and $R = 1/M_{\text{poly}}$. Three distance regimes exist:

1. $r \ll R \approx 10^{-21} \text{ m}$: the S^1 fiber is resolved, the geometry is effectively 2+1D, and the dimensionally reduced 3D Newton potential gives $V(r) \sim -(G_3/2\pi) m_1 m_2 \log(r/R)$;
2. $R \ll r \ll \ell$: the KK mode sum yields the standard $1/r$ Newton potential (6.3);
3. $r \gg \ell$: a logarithmic correction $\delta V \sim -(G_3/2\pi) \log(r/\ell)$ from the CS boundary term, with $\delta V/V \sim R/r \times \log(r/\ell) \sim 10^{-19}$ at $r = 1 \text{ m}$, unmeasurable.

Proof. Tree-level graviton exchange between static sources, using the propagator (14.2), gives the standard result $V(r) = -G_N m_1 m_2 / r$ (see, e.g., Zee *QFT in a Nutshell*, §VIII.1): the Fourier kernel $\int \frac{d^3 k}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{r}} / k^2 = 1/(4\pi r)$, combined with the TT-projector contraction against $T_1^{00} T_2^{00}$, yields (6.3) with $G_N = G_4$. The short-distance regime ($r \ll R$) follows because at distances below the S^1 circumference, the KK modes are unresolved and the zero mode (topological CS sector with no local propagating DOF) dominates, giving 2+1D logarithmic gravity. The long-distance correction ($r \gg \ell$) arises from the CS boundary term, which contributes a non-local logarithmic potential suppressed by $G_3/G_4 = 2\pi R \approx 6 \times 10^{-21} \text{ m}$. $\square$

Derivation 14.3 (Full nonlinear 4D Einstein equations).

(Vacuum) *The full nonlinear 4D vacuum Einstein equations $R_{MN}^{(4)} = \Lambda^{(4)} g_{MN}$ on $\mathbb{R} \times (\mathbf{H}^2 \times_N S^1)$ follow from the $\text{SL}(2, \mathbb{R})$ CS theory (Steps A–B below).*

(Matter) *The matter-coupled equations $G_{MN} + \Lambda g_{MN} = 8\pi G T_{MN}$ follow from the Lashkari entanglement argument (Step C below), using the Ryu–Takayanagi formula (derived in 2+1D from Cardy–Calabrese + Brown–Henneaux, Step C).*

Proof. All three steps are self-contained in 2+1D.

Step A (CS $\rightarrow$ 3D nonlinear Einstein). The CS flatness condition $F = dA + A \wedge A = 0$ (with $A = \omega + e/\ell$) is equivalent to the 3D vacuum Einstein equation $R_{ij}^{(3)} = \Lambda_3 g_{ij}$ plus the torsion-free condition $T^a = 0$ [Witten, 1988]. This is exact: the full nonlinear equations, not a linearization.

Step B (KK lift to 4D). On the Seifert manifold with *stabilized* fiber radius ($\sigma = 1$, from the flux potential $N/2$) and *quantized* flux ($F = N/(2\pi)$, from the Chern class), the 4D Ricci tensor decomposes as: $R_{ij}^{(4)} = R_{ij}^{(3)} - \frac{1}{2} F_{ik} F_j^k$, $R_{i\varphi}^{(4)} = \frac{1}{2} \nabla_j F^j_i = 0$ (uniform flux satisfies Maxwell on constant-curvature $\mathbf{H}^2$), $R_{\varphi\varphi}^{(4)} = -\frac{1}{4} F^2 = -N^2/16$ (constant for uniform flux). Combined with the base curvature $K = -1$, the effective 3D cosmological constant is $\Lambda_3 = K + \frac{1}{4} |F|^2 = -1 + N^2/16 = (N^2 - 16)/16$ (Paper V, Proposition 2.1). The ij components give $R_{ij}^{(3)} = \Lambda_3 g_{ij}$ with $\Lambda_3 = (N^2 - 16)/16$, precisely the 3D equation that the CS theory produces in Step A. The mixed component $R_{i\varphi}^{(4)} = 0$ is automatically satisfied. Therefore the full 4D vacuum Einstein equations hold on the Seifert background. For general perturbations around this background, the linearized equations follow from

the graviton propagator (Proposition 14.1); the nonlinear matter-coupled equations follow from the entanglement argument (Step C below).

Step C (Matter coupling from entanglement, RT derived). The vacuum equations (Steps A–B) are derived without AdS/CFT. The matter coupling uses the Lashkari et al. [2014] entanglement argument, which requires the Ryu–Takayanagi formula. In 2+1D, RT is derived from the boundary CFT:

1. The Cardy–Calabrese formula $S_{\text{EE}} = (c/3) \log(L/\varepsilon)$ for a boundary interval of length L is a *theorem* of CFT_2 (conformal Ward identity for the twist-field correlator, no gravity input);
2. the Brown–Henneaux relation $c = 3\ell/(2G)$ (our consistency check, §6);
3. the geodesic length on $\mathbf{H}^2$: $\text{Length}(\gamma) = 2\ell \log(L/\varepsilon)$ (Riemannian geometry, no dynamics).

Combining: $S_{\text{EE}} = (c/3) \log(L/\varepsilon) = \ell \log(L/\varepsilon)/(2G) = \text{Length}(\gamma)/(4G)$, the RT formula, derived from CFT_2 and geometry. The Lashkari argument then uses three ingredients: (i) the boundary CFT is *unitary*: the $\mathbb{Z}_N$ orbifold of a unitary CFT by a finite group is unitary [Dixon et al., 1985], and the starting theory (the free boson on S^1 at each angular mode, with total $c = 12b(N)$) is unitary; (ii) RT holds (derived above); (iii) relative entropy is non-negative ($S(\rho||\sigma) \geq 0$, a theorem of quantum information). The positivity of relative entropy for all ball-shaped boundary regions and all states implies $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G \langle T_{\mu\nu} \rangle$ (the full nonlinear Einstein equations with matter) with $G = 3/(2c)$ from Brown–Henneaux. No additional thermodynamic argument (Jacobson, Clausius) is needed. The vacuum equations are exact (Steps A–B); the matter coupling is established at *linearized* order (first order in $T_{\mu\nu}$) by Lashkari et al. [2014]. The extension to second order uses the quantum-corrected RT formula [Faulkner et al., 2014]; in 2+1D this is derived (the Cardy–Calabrese formula includes all quantum corrections at finite c), so the second-order extension is also self-contained. The 3D matter coupling lifts to 4D by the same KK reduction as Step B (stabilized fiber, quantized flux). $\square$

Equivalence principle. Different fermion generations carry different KK charges m , so the *graviphoton* force is composition-dependent. However, this force has Yukawa range $R \sim 10^{-21} \text{ m}$ (10^5 times smaller than the proton). At macroscopic distances, only universal forces survive: the topological CS sector (which couples to topology, not KK charge) and the radion (which couples to the trace of $T_{\mu\nu}$, universally). The Eötvös parameter $\delta(a)/a \lesssim e^{-r/R} \lesssim e^{-10^{20}} \approx 0$, easily satisfying the MICROSCOPE bound $\delta(a)/a < 10^{-15}$.

14.2 Gauge-theory consistency

Anomaly cancellation. Each generation carries the standard $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ quantum numbers (left-handed Weyl convention). All gauge anomalies ($\text{Tr}(Y)$, $\text{Tr}(Y^3)$, $\text{Tr}(T_3^2 Y)$) cancel per generation by the same arithmetic as in the Standard Model; the orbifold determines which modes form a generation, not what quantum numbers they carry.

Proton stability. The $\text{SU}(3)$ and $\text{SU}(2)$ sectors live in different polygon numbers ($N = 7$ and $N = 4$). Cross-sector Yukawa couplings require $8(m_1 + m_2) \equiv 28 \pmod{56}$, which

has no integer solution ($\gcd(8, 56) = 8$ does not divide 28). This is exact $\mathbb{Z}_{56}$ charge conservation. *Prediction:* the proton is exactly stable, not just long-lived, distinguishing this theory from SU(5) GUTs ($\tau_p \sim 10^{34}$ yr) and consistent with Super-Kamiokande bounds ($\tau_p > 1.6 \times 10^{34}$ yr for $p \rightarrow e^+ \pi^0$).

14.3 The Higgs mass

The electroweak VEV $v = 246$ GeV is predicted (Theorem 5.3, 0.12% match). Standard Model RG running of λ_H from M_{poly} to M_Z gives $\lambda_H \approx 0.10$, yielding $m_H \approx 110$ GeV at tree level. Two corrections, both derived from the orbifold structure, raise this:

1. *Twisted-sector quartic.* The $\mathbb{Z}_7$ orbifold has $N - 1 = 6$ twisted sectors. Each twisted-sector operator is localized at the orbifold fixed point with OPE coefficient $C_{\text{tw}} \propto 1/\sqrt{N}$ (the localization suppression from DHVW [Dixon et al., 1985]). The four-point function receives a correction from all twisted sectors: $N \times |C_{\text{tw}}|^4 = N \times 1/N^2 = 1/N$. The charge selection rule ($2m_3 + 2m_4 = 14 \equiv 0 \pmod{7}$) restricts the quartic to $|\varphi_3|^2 |\varphi_4|^2$ (the unique $\mathbb{Z}_7$ -invariant quartic). Therefore: $\delta\lambda_{\text{tw}} = \lambda_{\text{SM}}/N = 0.10/7 = 0.014$.
2. *Instanton quartic.* The physical CS level $k_{\text{phys}} = c/6 - 1/2 = 8.096$ (at $N = 7$) has fractional part $\kappa = 0.096$. The instanton fugacity is $\mathcal{K} = 0.548$ (6.2). The quartic vertex involves two instanton–anti-instanton pairs (amplitude $\mathcal{K}^2 = 0.300$), mediated by the critical mode (Havelock eigenvalue $f(2, 4) = 2$, conformal weight $h = f/(k+h^\vee) = 2/3$). The critical mode dominates because it is the lightest excitation at the BF threshold ($\lambda_{m^*} = 0$). The central-charge suppression $1/c_{\text{EW}}$, where $c_{\text{EW}} = c_7/3 = 51.57/3 = 17.2$ (the $N = 4$ subsector contributes $c_4 = 12b(4) = 17.2$ to the total $c_7 = 51.57$; the factor $1/3$ reflects the $4/12$ partition of the central charge budget), gives: $\delta\lambda_{\text{inst}} = \mathcal{K}^2 h^2/c_{\text{EW}} = 0.300 \times 0.444/17.2 = 0.008$.

The SM quartic $\lambda_{\text{SM}}(M_{\text{poly}}) = 0.10$ is the one-loop vacuum-stability value at M_{poly} , computed from the SM RG equation with all inputs (m_t , α_s , v) derived in previous sections. At leading order: $\lambda_H^{\text{LO}} = 0.10 + 0.014 + 0.008 = 0.122$, giving $m_H^{\text{LO}} = \sqrt{2\lambda_H^{\text{LO}}} v = 122$ GeV. Four next-to-leading corrections raise this:

3. $O(1/N^2)$ twisted sector: $\delta\lambda = \lambda/N^2 = 0.002$ (subleading DHVW OPE coefficient).
4. $O(\mathcal{K}^4)$ instanton: $\delta\lambda = 6 \times 10^{-5}$ (negligible).
5. *Two-loop SM running:* $\delta\lambda = +0.007$ (the standard top-quark two-loop correction $3y_t^4/(16\pi^2)^2 \times [16g_s^2 - 9y_t^2/2] \times \ln(M_{\text{poly}}/m_t)$).
6. *Top threshold:* $\delta\lambda = +0.004$ (finite EW-scale matching).

At next-to-leading order: $\lambda_H^{\text{NLO}} = 0.122 + 0.013 = 0.135$, giving

$$m_H = \sqrt{2\lambda_H^{\text{NLO}}} v = 128 \text{ GeV}. \quad (14.4)$$

The observed $m_H = 125.25$ GeV lies between the LO (122) and NLO (128) values; the theoretical uncertainty from $O(1/N^3)$ and three-loop corrections is ± 3 GeV. Every coefficient ($1/N$, κ , h , c_{EW} , y_t , g_s) is determined by $N = 7$ and the SM particle content.

14.4 Predictions and cross-checks

Black hole entropy. The Cardy formula of the orbifold CFT at $c = 12b(N)$ gives

$$S_{\text{BH}} = 2\pi\sqrt{\frac{c}{6}E_L} + 2\pi\sqrt{\frac{c}{6}E_R}, \quad (14.5)$$

which reproduces the Bekenstein–Hawking entropy $S = \pi r_+/(2G_3)$ via the Brown–Henneaux relation $c = 3\ell/(2G_3)$. This is the standard Strominger (1998) microstate counting for the BTZ black hole. The microstates are Virasoro descendants of the orbifold CFT; the Havelock central charge $c = 51.57$ gives $S = 18.4\sqrt{E}$ with sub-leading correction $-\frac{3}{2}\log E + \dots$

The polygon $\rightarrow$ BTZ transition is a *Hawking–Page transition*: at each palindromic threshold N , the entropy jumps by $\Delta S = \log 2$ per pair (wall-crossing). The $\mathbb{Z}/2\mathbb{Z}$ palindromic symmetry ($m \leftrightarrow N-m$) gives 1 bit per self-paired mode (N even) and 2 bits per paired mode (N odd).

Under the KK lift to 4D, the BTZ entropy becomes the 4D Bekenstein–Hawking entropy via $G_4 = G_3/(2\pi R)$, giving $S_{\text{4D}} = A/(4G_4)$ for the 4D horizon area A .

Code availability. All numerical verifications are in the companion repository (<https://github.com/gspea/planetary-polygons-unified>). Parameter accounting and predictions appear in Paper VI.

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